A Spatial–Spectral Denoising Algorithm for Accurate Temperature and Emissivity Retrieval From Airborne Thermal Infrared Hyperspectral Data

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***Abstract*—Airborne thermal infrared (TIR) hyperspectral data, with high spatial and spectral resolutions, enable the precise retrieval of land surface temperature (LST) and land surface emissivity (LSE) data for environmental monitoring and natural resource surveys. However, these sensors often suffer from increased noise due to platform and instrument factors. This study proposes an innovative spatial–spectral denoising algorithm for accurate temperature and emissivity retrieval (SSDATER), which includes: 1) applying unsupervised classification and spatial aggregation to the at-sensor radiance to reduce noise and using the emissivity spectral smoothness-constrained (ESSC) algorithm to estimate precise atmospheric parameters based on spatially aggregated observational data and 2) performing spectral aggregation on the at-sensor radiance to reduce noise. The spectrally aggregated data are then atmospherically corrected on the basis of the previously estimated atmospheric parameters, and finally, the temperature and emissivity separation (TES) algorithm is employed to retrieve accurate LST&E values. Assuming that autocorrelation exists in the LSE spectrum, principal component analysis (PCA)-based regression is used to convert the retrieved multispectral LSE into hyperspectral LSE. The SSDATER method was evaluated on simulated airborne Hyper-CAM LW data. The results show that the atmospheric parameters estimated by the SSDATER method were more accurate than those estimated from Hyper-CAM LW data using the ESSC algorithm and those simulated directly from reanalysis atmospheric profiles using the radiative transfer model. When identical atmospheric parameters were used, the SSDATER method achieved the highest accuracy for LST&E retrieval (LST RMSE: 1.01 K and average LSE RMSE: 0.016), thus outperforming the TES algorithm on original Hyper-CAM LW data (LST RMSE: 1.91 K and average LSE RMSE: 0.036). In summary, the proposed SSDATER method demonstrates the potential for high-precision LST&E retrieval from noisy airborne TIR hyperspectral data within a simulation-based validation framework, thereby supporting further application studies with real airborne observations.**

***Index Terms*—Airborne thermal infrared (TIR) hyperspectral, Hyper-CAM LW, temperature and emissivity separation (TES), thermal airborne hyperspectral imager (TASI).**

I. Introduction

LAND surface temperature (LST) and land surface emissivity (LSE) are fundamental parameters for climate, hydrological, ecological, and biogeochemical models [1], [2], [3], [4]. Accurate retrieval of LST&E data is essential for studies of energy and water balance, soil moisture, thermal inertia, and many other Earth and environmental science applications [5], [6], [7], [8].

The advancement of thermal infrared (TIR) hyperspectral remote sensing presents new opportunities and challenges for retrieving LST and hyperspectral LSE [9], [10], [11], [12], [13]. Satellite-based TIR hyperspectral instruments, such as AIRS/Aqua, IASI/MetOp-C, and HIRAS/FY-3-D, have been widely applied in atmospheric vertical profile retrieval and climate model assimilation [14], [15], [16]. However, their coarse spatial resolution (usually greater than 10 km) and inability to retrieve surface information beneath clouds limit their effectiveness for detailed surface studies, such as mineral exploration and urban analysis. In contrast, airborne TIR hyperspectral instruments operate at lower altitudes, offer much higher spatial resolution, and are less affected by clouds. These advantages make them invaluable for monitoring urban heat islands, assessing farmland soil moisture, and resource mapping [17], [18], [19], [20], [21], [22], [23], [24].

Prior knowledge of LSE is an essential input for many LST retrieval methods, such as the split-window algorithm [25]. However, owing to the high spatial resolution and dense spectral information of airborne TIR hyperspectral data, obtaining accurate LSE values through traditional means is often difficult, especially since some approaches rely on auxiliary visible or near-infrared data, which may not always be available or sufficiently precise. As a result, joint retrieval of LST and LSE has become essential. Many temperature–emissivity separation methods have been developed, including the iterative spectrally smooth temperature/emissivity separation (ISSTES) algorithm [26], [27], stepwise refining algorithm (SRTES) [13], linear spectral emissivity constraint (LSEC) algorithm [28],

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temperature and emissivity separation (TES) algorithm [29], multiscale wavelet-based TES algorithm (MSWTES) [12], subspace-based TES (SBTES) algorithm [30]. The accuracy of these temperature–emissivity separation methods, however, crucially depends on the quality of atmospheric correction [8], [31].

Atmospheric correction methods for airborne TIR hyperspectral data can be classified into two main categories: in-scene information-based methods and radiative transfer-based methods. The in-scene information-based methods include: in-scene atmospheric compensation (ISAC) algorithm [32], automatic atmospheric compensation (AAC) algorithm [33], emissivity spectral smoothness-constrained (ESSC) algorithm [34]. The ISAC algorithm derives atmospheric transmittance and upward radiance spectra directly from the hyperspectral image data by fitting lines to the top edge of radiance versus Planck function scatterplots, relying on the presence of near-blackbody surfaces and temperature variations in the scene, without requiring external atmospheric data. The AAC algorithm assumes that the LSEs of two adjacent hyperspectral bands (namely, an absorption band and a window band) are identical. It estimates the atmospheric parameters for all bands based on the scaling factor and intercept obtained from the regression of brightness temperatures between these two bands, which correspond to the ratio of atmospheric transmittances and the difference in path radiances, respectively. The ESSC algorithm first creates a transmission, upward radiance, downward radiance (TUD) database under varying conditions, then selects candidate atmospheric parameters from the TUD database on the basis of the initial guess, and performs atmospheric correction and temperature–emissivity separation with candidate atmospheric parameters, respectively. The optimal atmospheric parameters are identified by evaluating the smoothness of the retrieved LSE spectrum. Radiative transfer-based methods rely on atmospheric profiles (e.g., radiosonde or reanalysis data) input into radiative transfer models to calculate atmospheric parameters. Despite their utility, these approaches exhibit significant limitations. Some key assumptions of the ISAC algorithm, such as the existence of ideal blackbody surfaces, are often not met in practice. Moreover, because aircraft operate within the atmosphere, both ISAC and AAC algorithms are unable to accurately estimate atmospheric downward radiance solely from image data. The ESSC algorithm is sensitive to instrument noise and initial atmospheric parameters. In contrast, radiative transfer-based methods require highly accurate atmospheric profiles, which are rarely available with sufficient precision during airborne campaigns.

To address the limitations of separate atmospheric correction and surface parameter retrieval, integrated methods have been proposed. These methods, using physics-based or machine learning-based methods, directly retrieve both surface and atmospheric parameters from at-sensor radiance. For instance, Ye et al. [35] proposed a deep neural network (DNN) model combining a long short-term memory (LSTM) network and a convolutional neural network (CNN) to simultaneously estimate LST&E and atmospheric parameters from thermal airborne hyperspectral imager (TASI) observations, achieving an LST estimate accuracy of 0.95 K and an emissivity estimate accuracy of less than 0.012 for each channel. Nevertheless, machine learning-based integrated methods struggle with diverse atmospheric conditions and instrument types. Models trained on simulated data often fail to perform reliably on real measurements, as training parameters rarely capture the full range of real-world scenarios. Building upon this foundation, some researchers use land surface and atmospheric parameters obtained via machine learning as initial values, subsequently refining the retrieval results using physically based methods. For instance, Ren et al. used a DNN model to obtain initial estimates of land surface and atmospheric parameters from the TASI data, and then the initial estimates were input into a linearized radiative transfer model for further iterative optimization. Compared with the machine learning-based model, the physics-based model effectively improved the LST&E retrieval accuracy, reducing the LST RMSE from 0.98 to 0.86 K and achieving RMSEs below 0.015 for emissivity in all bands [36]. Despite the high accuracy offered by physics-based methods, they suffer from high computational complexity, strong dependence on prior knowledge, and may produce unstable retrieval results, limiting their practical application in operational environments.

Airborne TIR hyperspectral sensors are typically subject to greater instrument noise due to factors such as platform stability, integration time, background noise, anti-vibration compensation, and high spectral and spatial resolutions. Taking the widely used commercial TIR hyperspectral instrument Hyper-CAM LW as an example, its typical instrument noise-equivalent delta temperature (NEDT) is more than 0.5 K over most of the spectral bands [37]. Instrument noise substantially degrades the accuracy of the derived LST&E [38], [39]. Although various noise reduction techniques based on statistical approaches, signal processing algorithms, and modeling theories can effectively mitigate noise in at-sensor radiance measurements, these methods often risk altering the original physical characteristics of the data, potentially obscuring meaningful spectral information [40], [41], [42]. Therefore, a method that effectively minimizes noise in airborne TIR hyperspectral data while preserving the physical significance of the observations is both practically valuable and holds great promise.

This study proposes a spatial–spectral denoising algorithm for accurate temperature and emissivity retrieval (SSDATER) from airborne TIR hyperspectral data. The SSDATER method employs a dual aggregation strategy that suppresses noise while preserving the physical authenticity of the observational data by selectively trading spatial and spectral resolution at different stages. During spatial aggregation, it fuses information from spectrally consistent neighboring pixels, modestly lowering spatial resolution while preserving full spectral resolution to better exploit hyperspectral information for atmospheric parameter estimation. During spectral aggregation, it averages adjacent, highly correlated spectral channels within each pixel, lowering spectral resolution while preserving spatial resolution and at-sensor radiance, thus maintaining the image's spatial advantages and supporting precise temperature and emissivity retrieval. Consequently, the proposed approach shows promising potential to support accurate LST&E retrieval from airborne TIR hyperspectral data even in the presence of considerable instrument noise. The remainder of this article is organized as follows. Section II introduces the characteristics of Hyper-CAM LW and the simulation of its data; Section III provides a detailed description of the SSDATER method;

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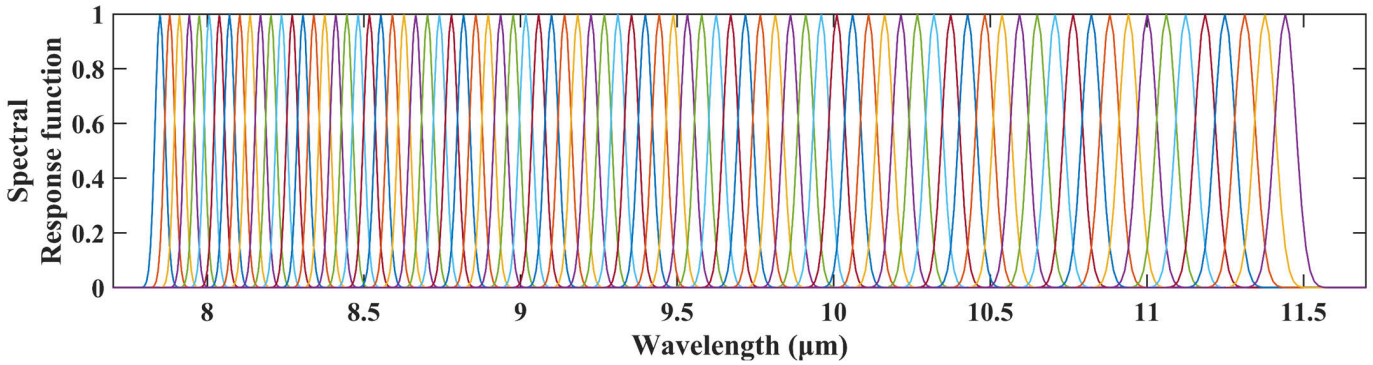


Fig. 1. Spectral response functions of the 81 bands of Hyper-CAM LW simulated via Gaussian functions.

TABLE I
CHARACTERISTICS OF THE HYPER-CAM LW INSTRUMENT

Specification	Value
Spectral range	7.87-11.45 μm
Spectral resolution	0.25 to 150 cm^{-1} (user adjustable)
Radiometric accuracy	<1 K
Typical noise equivalent spectral radiance (NESR)	< 20 $nW/cm^2/sr/cm^{-1}$

Section IV presents the validation results for atmospheric parameters, LST, and LSE; Section V discusses the results; and Section VI concludes the article by summarizing the key findings and insights.

II. DATA

As actual Hyper-Cam LW data were not available for this study, the effectiveness and reliability of the proposed SSDATER method were evaluated preliminarily via simulated airborne Hyper-CAM LW data. The at-sensor radiance of the airborne Hyper-CAM LW was simulated on the basis of the radiative transfer equation, with the surface parameters (i.e., LST&E) and atmospheric data derived from TASI products and corresponding radiosonde atmospheric profiles. Section II-A provides a brief introduction to the characteristics of the Hyper-CAM LW instrument, while Section II-B offers an overview of the simulation of at-sensor radiance for the airborne Hyper-CAM LW.

A. Characteristics of the Hyper-CAM LW

The Hyper-CAM LW is an airborne Fourier transform TIR hyperspectral imager that was developed by Telops Inc., Canada [43]. Its spectral range is 7.87–11.45 μm, with an adjustable spectral resolution that reaches 0.25 cm^{-1}. In addition, it is equipped with two internal blackbodies to ensure high-quality radiometric calibration. The Hyper-CAM LW sensor can be installed on an aircraft using its dedicated airborne module. The specifications of the Hyper-CAM LW sensor are summarized in Table I.

B. At-Sensor Radiance Simulation for the Airborne Hyper-CAM LW

Assuming that the land surface is a flat Lambertian surface, the spectral at-sensor radiance received by the airborne TIR sensor can be expressed by the following equation:

$$L_{TOA}(\lambda) = (\varepsilon(\lambda) \cdot B(T_s, \lambda) + (1 - \varepsilon(\lambda)) \cdot L_{\downarrow}(\lambda)) \cdot \tau(\lambda) + L_{\uparrow}(\lambda) \quad (1)$$

where $L_{TOA}(\lambda)$ represents the at-sensor radiance at wavelength λ; $\varepsilon(\lambda)$ and T_s represent the LSE and LST, respectively; $B()$ is the Planck function; $\tau(\lambda)$ and $L_{\uparrow}(\lambda)$ represent the atmospheric transmittance and the upward atmospheric radiance from the surface to the flight altitude, respectively; and $L_{\downarrow}(\lambda)$ represents the equivalent radiance of the downward hemispherical irradiance from the entire atmospheric column (hereafter referred to as atmospheric downward radiance).

Assuming that the spectral response function of each band of the Hyper-CAM LW is a Gaussian function, the theoretical at-sensor radiance of Hyper-CAM LW band i can be expressed as follows:

$$L_{TOA}(i) = \frac{\int_{\lambda_1}^{\lambda_2} L_{TOA}(\lambda) f(\lambda) d\lambda}{\int_{\lambda_1}^{\lambda_2} f(\lambda) d\lambda} \quad (2)$$

where $L_{TOA}(i)$ represents the radiance of the band i; λ_1 and λ_2 represent the wavelength range of band i; $f(\lambda)$ is the spectral response function of band i, which is simulated on the basis of a Gaussian function as follows:

$$f(\lambda) = \frac{1}{\sqrt{2\pi}\sigma_\lambda} \cdot \exp\left(-\frac{(\lambda - \lambda_0)^2}{2\sigma_\lambda^2}\right) \quad (3)$$

$$\sigma_\lambda = \frac{FWHM}{2\sqrt{2\ln 2}} \quad (4)$$

where λ_0 is the central wavelength, $FWHM$ is the full width at half maximum of band i. The spectral range of the Hyper-CAM LW is 7.87–11.45 μm, and the spectral resolution of the Hyper-CAM LW can be freely adjusted (within 0.25–150 cm^{-1}). Several studies have set the spectral resolution to 6 cm^{-1}, thus resulting in 81 spectral bands and a sampling interval of 5 cm^{-1}, to balance the spectral resolution and signal-to-noise ratio [44]. In this study, the same setting is adopted. Fig. 1 shows the simulated spectral response functions of the 81 bands. It is worth noting that while the theoretical equations are expressed in the wavelength domain (λ) for conceptual clarity, all actual forward simulations and spectral convolutions are implemented within the wavenumber domain (ν). Specifically, the central wavenumbers of the spectral response functions are set to be equally spaced at an interval of 5 cm^{-1}. This configuration yields 81 channels

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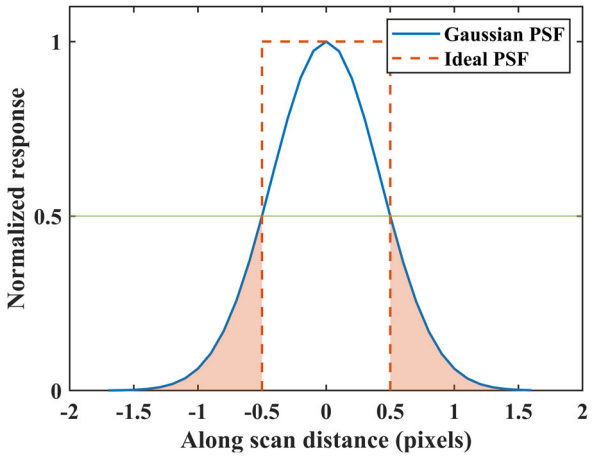


Fig. 2. Ideal and simulated PSFs (along the scan direction).

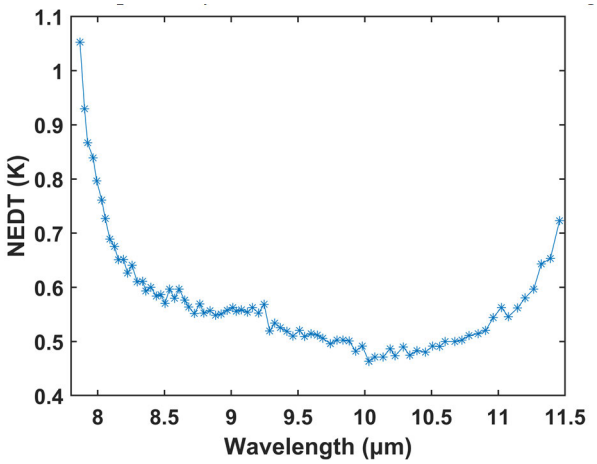


Fig. 3. NEDT across all 81 bands of Hyper-CAM LW.

covering the wavenumber range from 1274 to 874 cm^{-1}. To ensure energy conservation during convolution, the high-resolution reference spectra are converted from wavelength to the wavenumber grid using $L(\nu) = L(\lambda) \cdot 10^4/\nu^2$, where $L(\nu)$ is the spectral radiance in $W/cm^2/sr/cm^{-1}$, $L(\lambda)$ is the radiance in $W/cm^2/sr/\mu m$, and ν represents the wavenumber in cm^{-1}.

For a flat surface, the energy value of a pixel is entirely determined by the area that it covers under ideal conditions. However, in actual remote sensing images, a considerable portion of each pixel's spectral signal originates from its surrounding regions. This is due to a combination of factors, including the instrument's optical characteristics, detector and electronic components, atmospheric effects, and image resampling [45], [46], [47], [48]. Fig. 2 illustrates the spatial response of the sensor under ideal conditions and under "actual" conditions.

The actually observed at-sensor radiance can be expressed as the convolution of the ideal at-sensor radiance [i.e., (2)] with the sensor equivalent response function [i.e., point spread function (PSF)], together with the addition of a random noise process that represents the equivalent radiance noise [47]

$$L_{TOA,psf}(x,y,i) = \sum_{u=-k}^{k}\sum_{v=-k}^{k} L_{TOA}(x-u, y-v, i) \cdot PSF(u,v) + L_{noise}(x,y,i) \tag{5}$$

where x, y are the cross- and along-track coordinates of a pixel in the image space, while u, v are the cross- and along-track coordinates in the object space; $L_{TOA,psf}(x,y,i)$ is the at-sensor radiance of band i; $PSF(u,v)$ is the PSF; $L_{noise}(x,y,i)$ is the instrument noise or noise equivalent spectral radiance. In this study, the PSF of Hyper-CAM LW was modeled using a 2-D Gaussian function as follows:

$$PSF(u,v) = \frac{1}{2\pi\sigma_{psf}^2} \cdot \exp\left(-\frac{u^2+v^2}{2\sigma_{psf}^2}\right)$$
$$\sigma_{psf} = \frac{FWHM_{psf}}{2\sqrt{2\ln 2}} \tag{6}$$

where $FWHM_{psf}$ is the FWHM of the PSF, which was set equal to the spatial resolution of Hyper-CAM LW in this study.

Assuming that the instrument noise follows a Gaussian distribution, this study used a Gaussian distribution for simulation with the following equation, where $\sigma_{noise}(i)$ is the standard deviation of the noise-equivalent spectral radiance of band i:

$$L_{noise}(x,y,i) \sim N\left(0, \sigma_{noise}^2(i)\right). \tag{7}$$

These standard deviation values $\sigma_{noise}(i)$ were determined based on the NEDT specifications of the Hyper-CAM LW. The settings for NEDT at different bands for Hyper-CAM LW are primarily based on [44], as detailed in Fig. 3.

The at-sensor radiance of the airborne Hyper-CAM LW was simulated by (1), (2), and (5) with surface parameters (i.e., LST and LSE) and atmospheric data. The LST was derived from TASI data acquired during an airborne TIR hyperspectral remote sensing campaign that was conducted at 04:00 UTC on July 4, 2012, over the Linze experimental area and the middle reaches of the Heihe River in the Heihe River Basin [49]. The atmospheric profile data were obtained from radiosonde measurements at the Zhangye meteorological observatory during the TASI overpass. The LSEs were assigned on the basis of the land cover (LC) product obtained via unsupervised classification of the abovementioned TASI data. In the simulation, the flight altitude of the Hyper-CAM LW was set to 2.5 km. Details regarding the spectral response function, PSF, and instrument noise are provided above.

The spatial distributions of the LST and the LC type are shown in Fig. 4.

LSE values were assigned using the classification-based method, with spectra for each LC type selected from the ASTER spectral library [50] based on the abovementioned LC. The LSE spectrum for each LC type is shown in Fig. 5.

The radiosonde sounding profile was input into the moderate resolution atmospheric transmission (MODTRAN) model to calculate the atmospheric transmittance, upward radiance at a flight altitude of 2.5 km, and the downward radiance of the entire atmosphere. Fig. 6 shows the temperature and water vapor profiles, as well as the atmospheric parameters after convolution with the Hyper-CAM LW spectral response function.

Based on (1), (2), and (5), as well as the aforementioned surface and atmospheric parameters, this study simulated and generated the at-sensor radiance data of the Hyper-CAM LW. Fig. 7 shows the spatial distribution of the simulated at-sensor brightness temperature of the Hyper-CAM LW.

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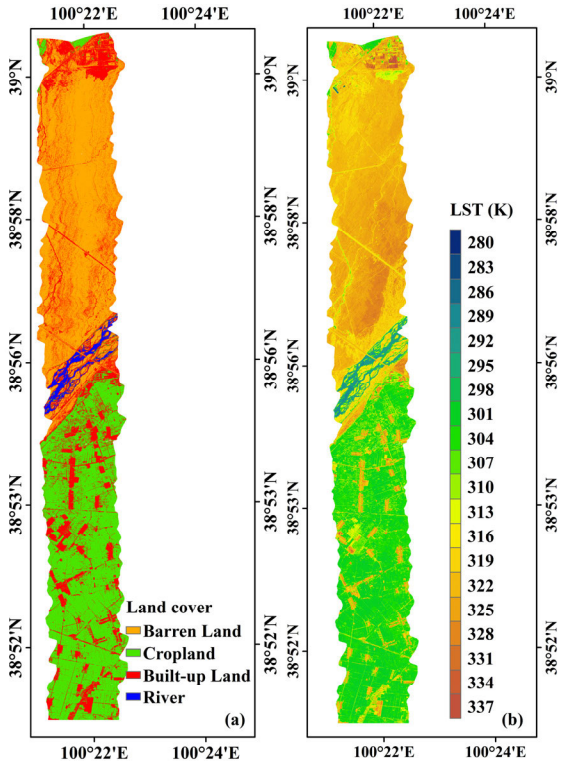


Fig. 4. Spatial distributions of the TASI LC (a) and LST (b).

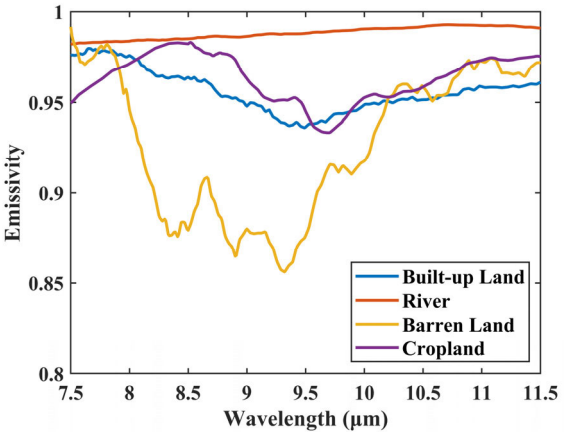


Fig. 5. LSE spectra that correspond to the four LC types.

As shown in Fig. 7, the brightness temperature of Band 1 was significantly lower than that of the other bands. This was mainly due to the relatively low atmospheric transmittance of Hyper-CAM LW Band 1. The simulated Hyper-CAM LW at-sensor radiance considers the PSF effect and instrument observation noise, which provides strong support for subsequent research on atmospheric correction and temperature–emissivity separation.

It should be noted that the LST, LSE, and atmospheric parameters constructed in this section are simulation-based references rather than independent observational ground truth. The TASI-derived LST, ASTER spectral-library-based emissivity assignment, radiosonde profile, and MODTRAN simulations provide internally consistent reference conditions for controlled algorithm evaluation. However, uncertainties

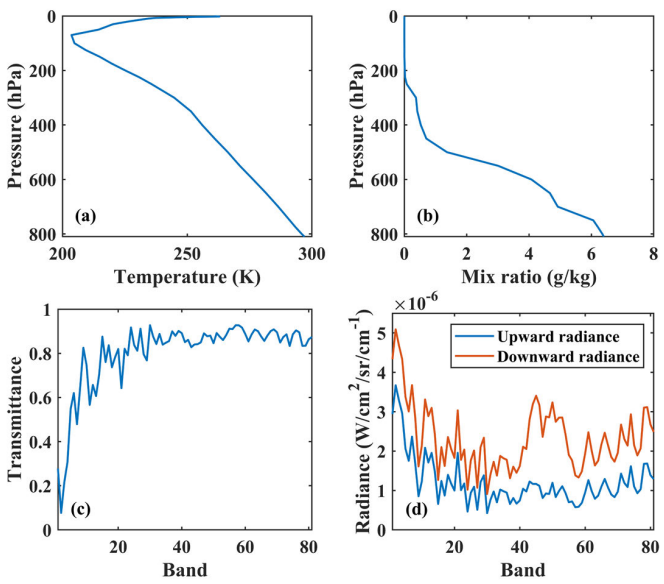


Fig. 6. Radiosonde profiles (a) and (b) at the time of TASI overpass and the corresponding atmospheric parameters (c) and (d).

from these sources are not fully propagated in the current work.

III. Methodology

As mentioned in the introduction, the accuracy of both integrated methods and temperature–emissivity separation methods decreases with increasing instrument noise. However, most existing denoising algorithms are primarily designed to enhance image sharpness and improve the signal-to-noise ratio, and their performance is usually assessed using image-quality metrics such as peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM). Nevertheless, most of these algorithms often struggle to fully maintain the spectral fidelity of denoised hyperspectral at-sensor radiance and may not deliver noticeable improvements in quantitative accuracy. Moreover, neither integrated algorithms nor temperature–emissivity separation methods explicitly incorporate a noise removal stage. To address this issue, we propose the SSDATER method for the high-precision retrieval of LST and LSE from Hyper-CAM LW data in the presence of instrument noise. The key innovation of the proposed SSDATER method is that, through spatial–spectral aggregation, it achieves multistage noise removal while preserving at-sensor radiance fidelity, thereby effectively supporting the accurate estimation of atmospheric parameters as well as LST&E.

To effectively reduce noise while preserving the physical significance of the observed data, SSDATER uses two strategies that specifically target the stages of atmospheric correction and temperature–emissivity separation as follows.

1) Assuming atmospheric uniformity over a large area, unsupervised classification and spatial aggregation (i.e., reduction of the spatial resolution over homogeneous surfaces) are iteratively applied to local blocks (e.g., 20×20 pixels) of the image data to lower the noise in the observed data and improve the accuracy of atmospheric correction using the ESSC algorithm.

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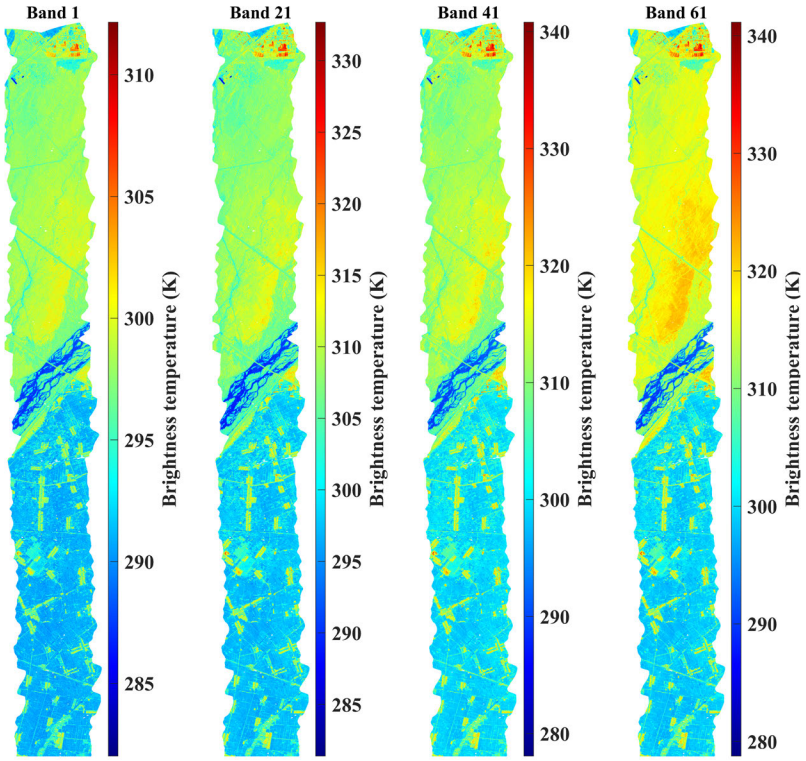


Fig. 7. Spatial distribution of the at-sensor brightness temperature of the Hyper-CAM LW at selected bands (Band 1: 7.85 μm, Band 21: 8.52 μm, Band 41: 9.31 μm, and Band 61: 10.27 μm).

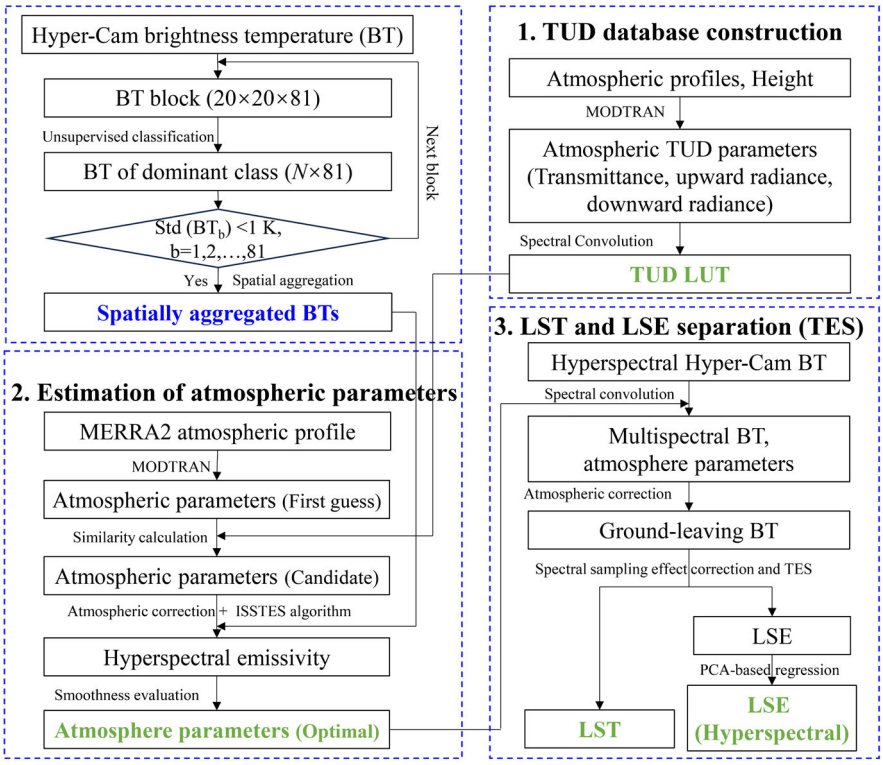


Fig. 8. Detailed flowchart of the SSDATER method proposed in this study.

2) Assuming that spectral autocorrelation exists for the LSE, spectral aggregation (i.e., reduction of the spectral resolution) is used to aggregate hyperspectral bands into several multispectral bands, thereby reducing noise and improving the accuracy of temperature–emissivity separation. In addition, a principal component analysis (PCA)-based regression method

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is employed to convert the retrieved multispectral LSE into hyperspectral LSE. Fig. 8 shows the flowchart of the developed SSDATER method.

Sections III-A and III-B introduce the atmospheric correction and the temperature–emissivity separation of the SSDATER method, respectively.

A. Atmospheric Correction

The original ESSC algorithm, by making use of the spectral information inherent in the image, overcomes the limitations of the ISAC and AAC algorithms in obtaining precise atmospheric downward radiation. However, the ESSC algorithm determines optimal atmospheric parameters by searching a range of candidates based on the smoothness of the LSE calculated by the ISSTES algorithm, making it sensitive to instrument noise. To address this issue, the SSDATER method proposed in this study offers an improvement over the ESSC algorithm, the atmospheric correction component of which consists of three components:

1) TUD Database Construction: The SeeBor V5.0 atmospheric profile dataset was employed to characterize the atmospheric state [51]. Marine atmospheric profiles were excluded to focus on terrestrial conditions. To mitigate cloud contamination, profiles with a relative humidity exceeding 90% at any pressure level were removed. This filtering process yielded a subset of 3517 terrestrial atmospheric profiles. The selected atmospheric profiles (i.e., air temperature, water vapor, and ozone profiles) were then input into the MODTRAN radiative transfer model to generate atmospheric transmittance, upward atmospheric radiance at the flight altitude (i.e., 2.5 km), and downward atmospheric radiance. These atmospheric parameters were convolved with the spectral response function of Hyper-CAM LW data to generate the TUD database.

2) Localized Unsupervised Clustering and Spatial Aggregation: Spatial aggregation can preserve the physical meaning of the at-sensor radiance while maintaining spectral resolution. However, since the surface is not homogeneous due to differences in LSE and LST, directly applying simple spatial aggregation may introduce additional errors. To address this, a new spatial aggregation strategy was proposed.

First, an $N \times N$ window is slid across the entire Hyper-CAM LW image space, and unsupervised classification based on the K-means clustering algorithm is performed on the N^2 pixels within each window. The feature vectors used for clustering are original at-sensor brightness temperature values from all 81 bands. No normalization, standardization, or band weighting was applied prior to clustering. A smaller window fails to suppress noise effectively, while an overly large window causes excessive within-class heterogeneity and violates the aggregation requirement. In practice, we used a 20×20 clustering window (i.e., $N = 20$) with a sliding stride of 20. Five clusters were determined empirically based on five typical LC types, namely bare soil, vegetation, water body, snow/ice, and artificial land. This configuration performs well for the proposed method and has been verified in practice.

Second, the most dominant class within the 20×20 window is analyzed. If the at-sensor brightness temperature standard deviation is below 1 K for each of the 81 bands, the dominant class is regarded as homogeneous. The pixels of this class are then averaged using a simple arithmetic mean to produce spatially aggregated at-sensor brightness temperature data. For Gaussian noise, this method serves as the optimal unbiased estimator. It is computationally simple, introduces no extra parameters, and fully retains the physical meaning of the data.

Third, steps 1 and 2 are repeated across the entire image. Fig. 9 illustrates an example of spatial aggregation.

As shown in Fig. 9(a), there is significant heterogeneity within the 20×20 pixel area. By applying unsupervised classification, pixels with similar spectral characteristics are grouped, which substantially reduces the within-class LST&E variation [Fig. 9(b)]. Spatial aggregation is then restricted to classes with a brightness temperature standard deviation below 1 K across all 81 spectral bands, which further minimizes uncertainties from surface heterogeneity. This approach improves the signal-to-noise ratio of at-sensor brightness temperature while effectively preserving atmospheric absorption features, as illustrated in Fig. 9(c).

3) Candidate Atmospheric Parameter Selection and Optimal Atmospheric Parameter Estimation: The reanalysis atmospheric profiles [e.g., Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA2)] are used as the background field and input into MODTRAN to calculate the initial values of atmospheric parameters. Then, the spectral angle between the transmittance data of the initial atmospheric parameters and the atmospheric transmittance data in the TUD database is computed

$$\theta = \cos^{-1}\left(\frac{A \cdot B}{\sqrt{\sum_{i=1}^{81} A_i^2} \cdot \sqrt{\sum_{i=1}^{81} B_i^2}}\right) \tag{8}$$

where A represents the transmittance at the 81 bands of the initial atmospheric parameters, and where B refers to the atmospheric transmittance data from the TUD database. θ is the spectral angle; the smaller its value is, the greater the similarity between the two transmittance spectra. On the basis of the calculated spectral angles, the 40 atmospheric parameters most similar to A are selected from the TUD database. This empirically determined number is set as a trade-off: a smaller number of candidates would result in selected profiles highly analogous to the background atmospheric field and insufficient diversity of atmospheric candidates, whereas excessive candidates will drastically increase computational burden with no substantial gain in retrieval precision.

Based on the selected candidate atmospheric parameters, atmospheric correction is applied to the spatially aggregated at-sensor radiance. Then, the ISSTES algorithm is used for LST and LSE retrieval based on the spatially aggregated at-sensor radiance. The smoothness of the retrieved LSE is then calculated by the following equation:

$$S = \sum_{i=8}^{80}\left(\varepsilon(i) - \frac{\varepsilon(i-1) + \varepsilon(i) + \varepsilon(i+1)}{3}\right) \tag{9}$$

where S represents the smoothness of the LSE spectrum. Due to strong atmospheric absorption in the first 10 bands, these channels are more susceptible to uncertainties in atmospheric correction and show relatively high NEDT values as illustrated in Fig. 3. Stepwise band elimination experiments demonstrate that removing the first six bands leads to a substantial improvement in retrieval accuracy, whereas further removing additional bands yields only negligible accuracy

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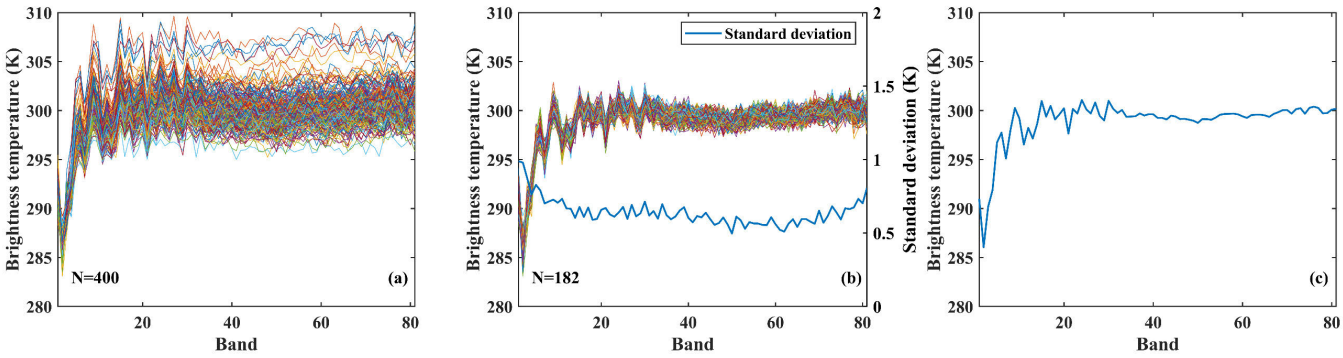


Fig. 9. Brightness temperatures of pixels within a 20 × 20 block (a), as well as the brightness temperatures of the dominant LC type after unsupervised classification (b) and their aggregated results (c).

gains relative to the overall retrieval uncertainty and may even degrade retrieval performance. We therefore excluded the first six bands from both the LSE calculation using the ISSTES algorithm and the smoothness calculation using (9). For each candidate atmospheric parameter, the smoothness of the resulting LSE is evaluated, and the parameter that yields the smoothest LSE (i.e., the lowest S value) is considered optimal for that pixel. Finally, after statistical analysis of all the spatially aggregated at-sensor radiance results, the atmospheric parameter with the most votes is regarded as the optimal parameter.

If the Hyper-CAM LW coverage area shows negligible topographic relief, the atmosphere over the entire scene can be assumed spatially uniform, and atmospheric parameters can thus be estimated directly using the above algorithm. For regions with significant terrain variation, the image can be segmented according to elevation. Atmospheric profiles are adjusted to local surface elevation by removing vertical levels below the terrain and interpolating the remaining profile to match the local surface and sensor altitude according to the corresponding elevation of each pixel. The proposed algorithm can then be applied separately to each group of Hyper-CAM LW pixels within regions of similar atmospheric conditions.

B. Temperature and Emissivity Separation

The TES algorithm is one of the most widely used temperature–emissivity separation methods, and it can be recalibrated and applied to both multispectral and hyperspectral data. In this study, the TES algorithm was intended to be used to retrieve LST&E from Hyper-CAM LW data. However, the accuracy of the TES method not only depends on precise atmospheric correction, but also is sensitive to instrument noise. Instrument noise can increase the LSE contrast, thereby leading to underestimation of the minimum LSE and overestimation of the LST. For the Hyper-CAM LW, the NEDT of most bands is above 0.5 K. Therefore, it is necessary to find an effective way to reduce instrument noise to guarantee accurate LST and LSE estimates.

Assuming that the noise in each band is independent and identically distributed Gaussian noise (with standard deviation σ), after averaging over N bands, the noise level (the standard deviation of the noise) reduces to $(1/\sqrt{N})$ of the original value. In addition, LSE possesses strong spectral autocorrelation and information redundancy. As demonstrated by Li et al. [52] using empirical orthogonal function (EOF) analysis, the emissivity spectrum can be sufficiently represented by a small number of basis components. Specifically, the first six components explain more than 98% of the total variance, enabling accurate reconstruction of hyperspectral LSE from just a few key bands [52]. Therefore, the SSDATER method employs a spectral aggregation approach to improve the signal-to-noise ratio by reducing spectral resolution without compromising spatial resolution.

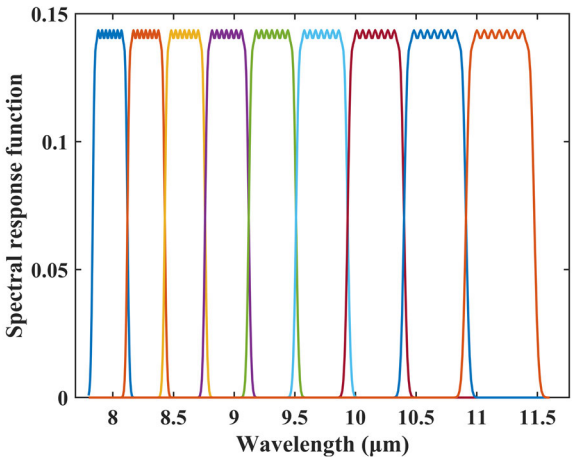


Fig. 10. Spectral response functions of the nine synthesized multispectral bands.

By applying spectral aggregation, the hyperspectral data are transformed into multispectral data. The previously obtained atmospheric parameters are then used to perform atmospheric correction, which results in the multispectral bottom-of-atmosphere (BOA) radiance. The TES algorithm is subsequently employed to retrieve the LST and multispectral LSE from these data. Finally, a PCA-based regression method is used to recover the hyperspectral LSE from the derived multispectral LSE.

In this study, we reduced the 81 hyperspectral bands of Hyper-CAM LW to nine synthetic multispectral bands; the corresponding spectral response functions of the nine synthetic bands became the following:

$$f(b) = \frac{1}{9}\sum_{i=1}^{9} f(i) \tag{10}$$

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TABLE II
CORRESPONDENCE BETWEEN THE 81 HYPER-CAM LW HYPERSPECTRAL BANDS AND NINE SYNTHETIC MULTISPECTRAL BANDS

Synthetic multispectral bands (Band b)	Hyperspectral bands of Hyper-CAM LW (Band i)
1	1, 2, 3 … 9
2	10, 11, 12 … 18
3	19, 20, 21 … 27
4	28, 29, 30 … 36
5	37, 38, 39 … 45
6	46, 47, 48 … 54
7	55, 56, 57 … 63
8	64, 65, 66 … 72
9	73, 74, 75 … 81

where $f(b)$ represents the spectral response functions of the nine synthetic multispectral bands ($b = 1, 2, 3, \ldots, 9$), which are expressed as the direct average of the spectral response functions of the hyperspectral bands ($f(i)$). This equal-weight scheme provides a fixed, transparent linear mapping that simplifies calibration procedures and subsequent regression between hyperspectral parameters and synthesized multispectral parameters (e.g., at-sensor radiance, atmospheric parameters). The correspondence between 81 hyperspectral bands of Hyper-CAM LW and nine synthetic multispectral bands is shown in Table II. The spectral response functions of the nine synthetic multispectral bands are shown in Fig. 10.

Since the spectral response function of each synthetic band is the numerical average of the spectral response functions of the corresponding nine hyperspectral bands, after spectral aggregation, the at-sensor radiance and atmospheric parameters of the synthetic bands can be expressed as follows:

$$y(b) = \sum_{i}^{N=9} k_i \cdot x_i + k_0 \tag{11}$$

where $y(b)$ represents the at-sensor radiance or atmospheric parameters of synthetic band b, and x_i refers to the corresponding parameters of the Hyper-CAM LW hyperspectral bands within the spectral range of each synthetic band. The k_i and k_0 are synthetic band-specific coefficients rather than atmosphere-specific coefficients. They can be obtained by inputting representative atmospheric and surface parameters (e.g., SeeBor atmospheric profiles and the ASTER emissivity spectral library) into MODTRAN to simulate the hyperspectral and multispectral at-sensor radiance of Hyper-CAM LW and the relevant atmospheric parameters, followed by least squares fitting. The constructed simulation dataset contains $3517 \times 89 \times 6$ samples, consisting of three input components: 3517 atmospheric profiles from the SeeBor database, 89 emissivity spectra sourced from the ASTER spectral library, and LST values derived from six independent perturbations applied to the near-surface air temperature of every single atmospheric profile. Based on the foregoing least-squares fitting using this entire constructed simulation dataset, the RMSE values for (11) converting at-sensor radiance, atmospheric transmittance, upward atmospheric radiance, and downward atmospheric radiance are 2.99×10^{-13} W/(cm^2·cm^{-1}·sr), 5.36×10^{-8}, 3.04×10^{-13} W/(cm^2·cm^{-1}·sr), and 5.84×10^{-13} W/(cm^2·cm^{-1}·sr), respectively. The coefficient of determination (R^2) exceeds 0.99 for all conversions. It should be emphasized that all conversion relationships derived from (11) are empirical approximations fit within the range of simulated atmospheric and surface conditions in this dataset. In addition, all RMSE and R^2 metrics presented above are internal fitting results calculated on the full simulation dataset, and no independent dataset is reserved for out-of-sample validation.

Overall, directly averaging adjacent bands for spectral aggregation has the following advantages: 1) adjacent bands exhibit a significant degree of similarity, both in terms of at-sensor radiance and spectral emissivity. As a result, aggregating these bands does not lead to a substantial loss of information; 2) the noise levels of adjacent bands are relatively consistent, aligning with the assumptions of spectral aggregation, where N bands have uncorrelated noise with similar amplitudes; and 3) the hyperspectral at-sensor radiance and atmospheric parameters can be converted to synthetic multispectral bands with minimal loss of accuracy.

However, after atmospheric correction using the at-sensor radiance and atmospheric parameters of the synthetic multispectral bands, the resulting BOA radiance [$BOA_{\text{est}}(b)$] and the true BOA radiance [$BOA_{true}(b)$] exhibit differences due to spectral sampling effects [53]. The differences between them are shown in the following equations:

$$BOA_{true}(b) = \int_{\lambda_1}^{\lambda_2} \frac{L_{TOA}(\lambda) - L_{\uparrow}(\lambda)}{\tau(\lambda)} \cdot f(\lambda) d\lambda \tag{12}$$

$$\begin{aligned} BOA_{\text{est}}(b) &= \frac{\int_{\lambda_1}^{\lambda_2} L_{TOA}(\lambda) \cdot f(\lambda) d\lambda - \int_{\lambda_1}^{\lambda_2} L_{\uparrow}(\lambda) \cdot f(\lambda) d\lambda}{\int_{\lambda_1}^{\lambda_2} \tau(\lambda) \cdot f(\lambda) d\lambda} \\ &= \frac{L_{TOA}(b) - L_{\uparrow}(b)}{\tau(b)} \end{aligned} \tag{13}$$

where $BOA_{true}(b)$ represents the true ground-leaving radiance of synthetic multispectral band b, $f(\lambda)$ is the normalized spectral response function of the synthesized multispectral band b, and $BOA_{est}(b)$ represents the estimated ground-leaving radiance obtained after atmospheric correction, which is calculated from the convolved at-sensor radiance, convolved upward radiance, and transmittance. Therefore, discrepancies exist between $BOA_{true}(b)$ and $BOA_{\text{est}}(b)$, and this inconsistency is especially pronounced in regions with significant differences in atmospheric absorption. We use Taylor expansion to approximate the relationship between the two

$$\begin{aligned} &BOA_{\text{true}}(b) \\ &\approx BOA_{\text{est}}(b) + \frac{L_{TOA}(b) - L_{\uparrow}(b)}{\tau(b)^3} \int (\tau(\lambda) - \tau(b))^2 \cdot f(\lambda) d\lambda \\ &\quad + \frac{1}{\tau(b)^2} \int (L_{\uparrow}(\lambda) - L_{\uparrow}(b) - L_{TOA}(\lambda) \\ &\quad + L_{TOA}(b)) \cdot (\tau(\lambda) - \tau(b)) \cdot f(\lambda) d\lambda \end{aligned} \tag{14}$$

where $L_{TOA}(b)$, $L_{\uparrow}(b)$, and $\tau(b)$ represent the at-sensor radiance, upward radiance, and transmittance, respectively, of synthetic multispectral band b; $f(\lambda)$ is the normalized spectral response function of band b; and $L_{TOA}(\lambda)$, $L_{\uparrow}(\lambda)$ and $\tau(\lambda)$ represent the at-sensor radiance, upward radiance, and transmittance of wavelength λ within the spectral region of synthetic multispectral band b, respectively. The detailed derivations of (14) can be found in the Appendix at the end of this article.

Using (14) to correct $BOA_{\text{est}}(b)$ can significantly mitigate errors that result from spectral sampling effects. However, since the at-sensor radiance of Hyper-CAM LW is not monochromatic [see (2)] and contains substantial noise, this

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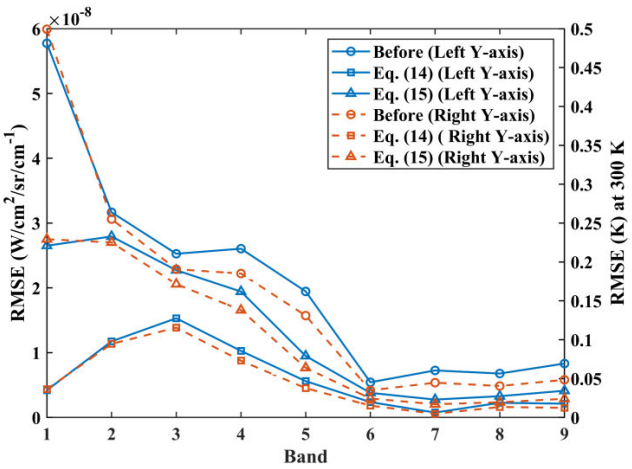


Fig. 11. Error in BOA radiance caused by the spectral sampling effect and after correction using two different methods. Results are shown in radiance units (left y-axis) and temperature units at 300 K (right y-axis).

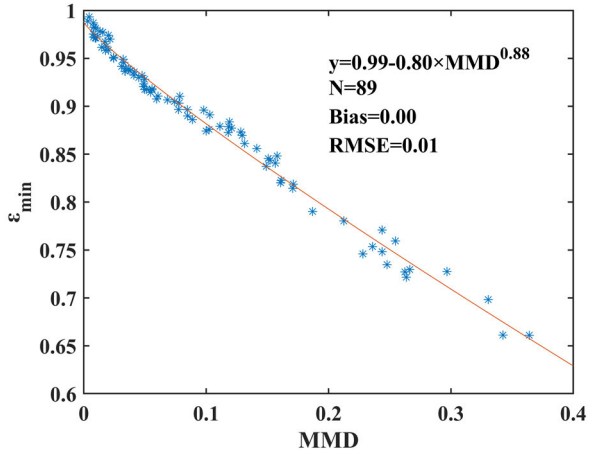


Fig. 12. Results of the TES empirical formulas recalibrated for the nine synthesized multispectral bands.

study uses the following equation for spectral sampling effect correction:

$$BOA_{true}(b) \approx k_1 \cdot BOA_{\text{est}}(b) + k_2 \cdot \frac{L_{TOA}(b) - L_{\uparrow}(b)}{\tau(b)^3} \cdot \sum_{i=1}^{9} (\tau(i) - \tau(b))^2 \quad (15)$$

where $\tau(i)$ represents the transmittance of the corresponding nine hyperspectral bands within band b (see Table II) and k_1 and k_2 are unknown constants that can be determined by simulating $BOA_{\text{est}}(b)$ and $BOA_{true}(b)$ as well as the corresponding atmospheric parameters, with representative atmospheric and surface values used in a radiative transfer model, followed by least squares fitting. Fig. 11 shows the efficacy of spectral sampling effect correction for $BOA_{\text{est}}(b)$ using (14) and (15).

As shown in Fig. 11, (14) achieves high theoretical accuracy and can effectively reduce errors caused by the spectral sampling effect. Although the theoretical accuracy of (15) is slightly lower than that of (14), it can still improve the accuracy of $BOA_{\text{est}}(b)$.

The TES algorithm can effectively separate LST and LSE. The empirical relationship between minimum LSE and LSE contrast in the TES algorithm must be recalibrated for each specific instrument. In this study, the TES empirical formula was recalibrated using the 89 emissivity spectra from the ASTER emissivity spectral library and the spectral response function of nine synthesized multispectral bands. The calibration results are presented in Fig. 12, and all fitting metrics are in-sample results derived from all 89 spectra, as no data were split for validation.

Since there is redundancy within the LSE spectrum, under acceptable levels of information loss, representing the spectrum using only a subset of principal components is possible [54]. On the basis of this idea, the SSDATER method uses a PCA-based regression method to recover the hyperspectral LSE from the multispectral LSE. The basic PCA regression equation can be expressed as follows:

$$\begin{aligned} \Delta Y^T &= C(\Delta X \cdot U_p)^T \\ \Delta X &= X - \bar{X} \\ \Delta Y &= Y - \bar{Y} \end{aligned} \quad (16)$$

where Y is the hyperspectral LSE spectrum matrix, $\bar{Y}$ is the mean value of Y, X is the multispectral LSE spectrum, $\bar{X}$ is the mean value of X, U_p represents the first n principal components of ΔX, and C is the coefficient matrix, which can be calculated as follows:

$$\begin{aligned} C &= \Delta Y^T \cdot A \cdot \left(A^T \cdot A\right)^{-1} \\ A &= (\Delta X \cdot U_P)^T. \end{aligned} \quad (17)$$

In this study, the ASTER emissivity spectral library was convolved with the spectral response functions of 81 Hyper-CAM LW bands and nine synthetic multispectral bands, respectively, to obtain hyperspectral and multispectral emissivities. The coefficient C can then be determined by (17), with fitting performed on all 89 emissivity spectra from the ASTER spectral library. By keeping the values of $\overline{X}$, $\overline{Y}$, U_P and C unchanged and letting X be the multispectral LSE (nine synthetic multispectral bands), the LSE spectra of Hyper-CAM LW (81 bands) which are denoted by Y can be derived from (16) and (17). Fig. 13 presents the reconstruction of hyperspectral LSE from the nine synthetic multispectral bands using the PCA-based regression method. To retain all information contained in the nine multispectral bands and minimize the reconstruction error, all nine principal components were adopted in the final regression model. As shown in Fig. 13(a) and (b), this configuration enables highly faithful recovery of Hyper-CAM LW emissivity spectra, and Fig. 13(c) further indicates that employing all nine principal components yields the lowest reconstruction RMSE, with the error across all 81 spectral bands remaining below 0.004. These statistics are in-sample results calculated from the full set of 89 ASTER emissivity spectra, which were also used to fit the PCA regression model; no independent test spectra were employed for validation. Notably, the nine principal components only preserve the information available in the nine-band multispectral vector rather than the full information originally contained in the 81-band hyperspectral spectrum.

IV. Results

A. Atmospheric Correction

In this study, 6-hourly MERRA2 reanalysis atmospheric profiles—including air temperature, water vapor, and

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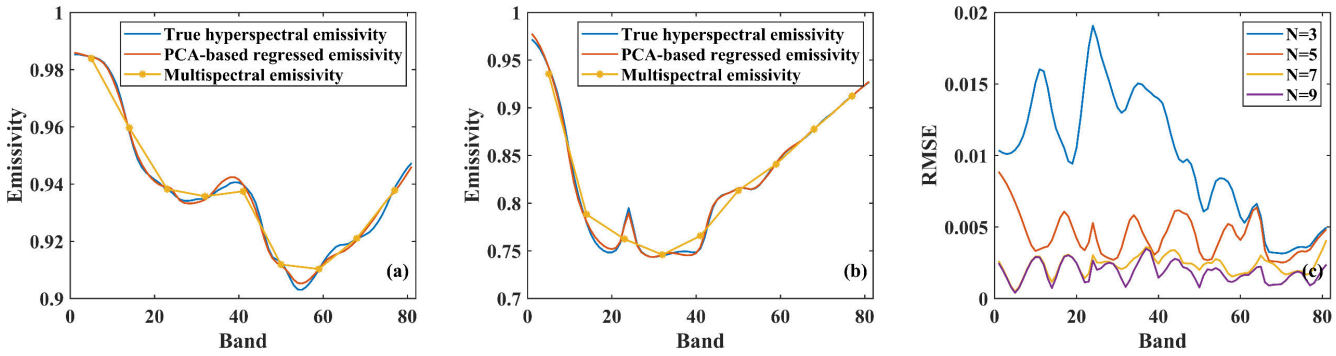


Fig. 13. Examples of hyperspectral LSE restoration from multispectral LSE using PCA regression with all nine components (a) and (b), and reconstruction RMSE for different numbers of principal components (c).

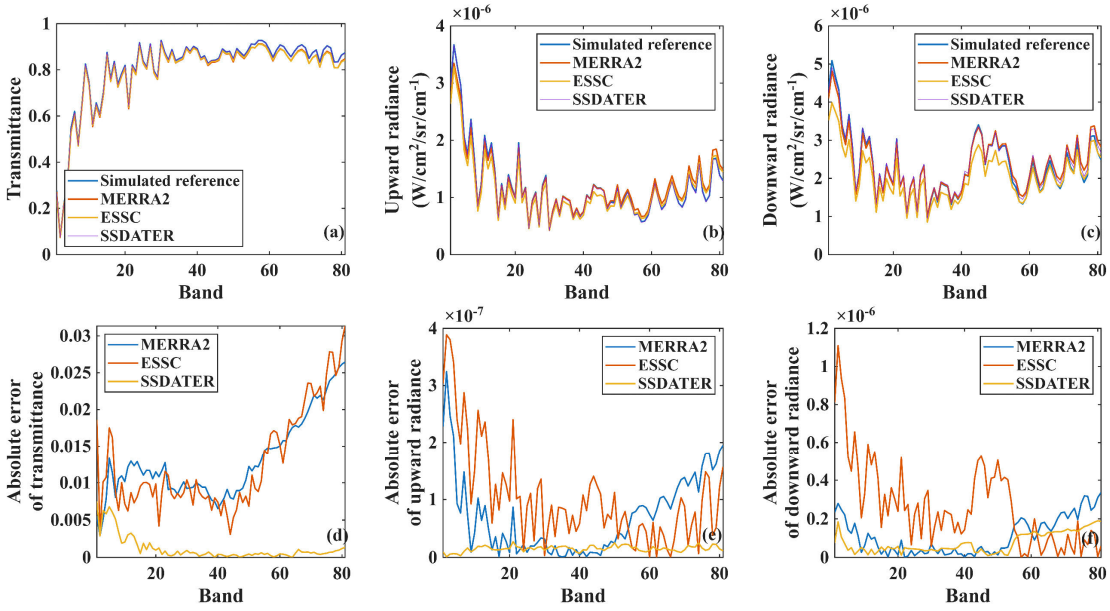


Fig. 14. Comparison of atmospheric parameters obtained using different methods with the simulated reference atmospheric correction (a)–(c), and their corresponding errors (d)–(f).

ozone—were used as background fields. First, the MERRA2 atmospheric profile at 04:00 UTC on July 4, 2012, was derived through linear interpolation between the two nearest available time points (00:00 UTC and 06:00 UTC on the same day). The simulated Hyper-CAM LW data consist of 5945 pixels in length and 864 pixels in width, with a spatial resolution of 3 m, producing a total swath of 17.84×2.59 km. Given that the spatial resolution of the MERRA2 reanalysis data is $0.625° \times 0.5°$, the entire Hyper-CAM LW scene is almost completely covered by a single MERRA2 grid cell. Therefore, the MERRA2 atmospheric profile corresponding to the geographic coordinates of the central pixel of the Hyper-CAM LW image was extracted using the nearest-neighbor method and employed in this study.

The interpolated MERRA2 atmospheric profiles were then input into MODTRAN to calculate the initial atmospheric parameters for Hyper-CAM LW. To demonstrate the effectiveness of the SSDATER method, we input the initial atmospheric parameters estimated from MERRA2 directly into the ESSC algorithm and SSDATER method to obtain the optimal atmospheric parameters. Fig. 14 shows the simulated reference atmospheric parameters, the atmospheric parameters estimated from MERRA2, and the optimal atmospheric parameters estimated by the ESSC algorithm and SSDATER method.

As shown in Fig. 14, based on the initial atmospheric parameter values estimated from MERRA2 and the original Hyper-CAM LW at-sensor radiance, the atmospheric parameters derived using the ESSC algorithm exhibit relatively poor accuracy, which is sometimes even worse than that of the initial values. The atmospheric parameters estimated by the SSDATER method are closer to the simulated reference atmospheric parameters and have higher accuracy, thus effectively supporting subsequent temperature–emissivity separation.

B. Temperature and Emissivity Separation

Based on the atmospheric parameters (81 bands) obtained in Section IV-A, the atmospheric parameters for the nine synthesized multispectral bands were derived by (11). The at-sensor radiance of these nine synthesized multispectral bands was subsequently corrected for atmospheric effects and

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TABLE III
DETAILED CONFIGURATION OF EACH GROUP

Group	Atmospheric parameters	Band of at-sensor radiance
Group 1	MERRA2	Hyper-CAM LW hyperspectral band (7, 8, 9 … 81)
Group 2	SSDATER	Hyper-CAM LW hyperspectral band (7, 8, 9 … 81)
Group 3	SSDATER	Synthesized multispectral band (1, 2, 3 … 9)

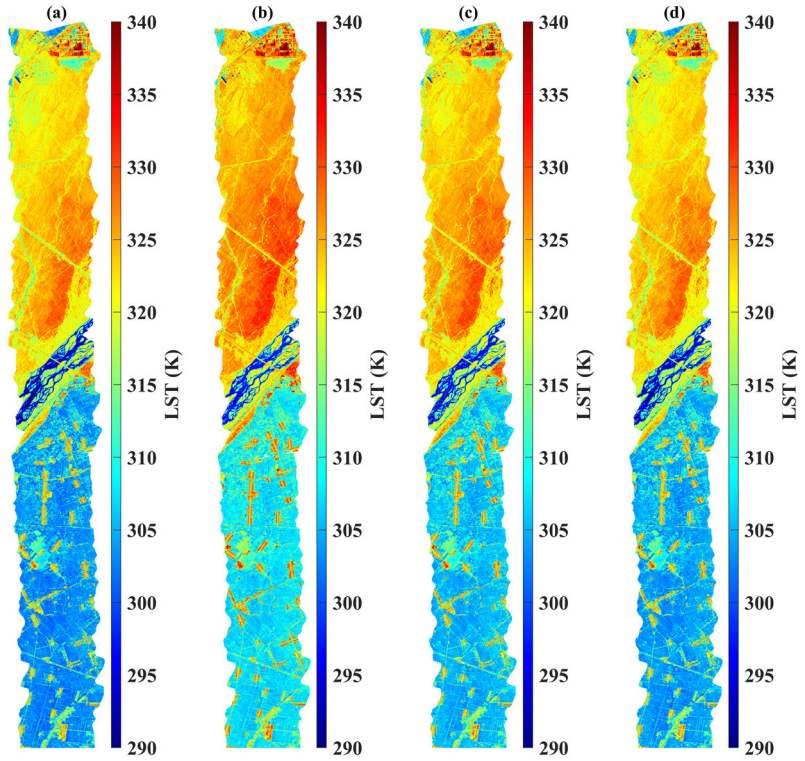


Fig. 15. Spatial distributions of the simulated reference LST (a), and LSTs retrieved from three groups: (b) Group 1, (c) Group 2, and (d) Group 3.

spectral sampling effects to obtain the BOA radiance. In this section, the TES algorithm was used to separate the LST and LSE, and the PCA-based regression method was applied to recover the hyperspectral LSE from the retrieved multispectral LSE.

To further assess the overall effectiveness of the SSDATER method, two sets of comparative experiments were conducted as follows.

1) Atmospheric correction was performed on the original Hyper-CAM LW at-sensor radiance data using both atmospheric parameters from MERRA2 and those estimated by the SSDATER method. The TES algorithm was subsequently applied to retrieve the LST&E. This experiment was used to further evaluate the effectiveness of the atmospheric parameters estimated by the SSDATER method.

2) Atmospheric parameters obtained through the SSDATER method were used to perform atmospheric correction on both the Hyper-CAM LW hyperspectral at-sensor radiance data and the nine synthesized multispectral at-sensor radiance data. Subsequently, the recalibrated TES algorithm was applied to retrieve LST&E from each dataset, respectively. The multispectral LSE was then further processed using a PCA-based regression method to reconstruct the hyperspectral LSE. This experiment aimed to evaluate the reliability of the high-accuracy retrieval of LST&E data using the SSDATER method.

The experiments were categorized into three distinct groups based on the varying atmospheric parameters and the number of spectral bands utilized. The detailed configurations of each group are presented in Table III. Owing to excessive noise in the first 6 bands of the original Hyper-CAM LW data, these bands were excluded during the temperature–emissivity separation for Group 1 and Group 2.

Fig. 15 shows the spatial distributions of the LSTs retrieved from Groups 1–3, as well as the simulated reference LST.

As shown in Fig. 15, the resulting LST values exhibited the largest deviation from the simulated reference values when atmospheric parameters from MERRA2 were used together with the original Hyper-CAM LW at-sensor radiance (i.e., Group 1) for LST retrieval, with significant overestimation in both cropland and bare soil areas. Replacing the atmospheric parameters with those obtained using the SSDATER method (i.e., Group 2) resulted in the spatial distributions of the retrieved LSTs being noticeably closer to the simulated

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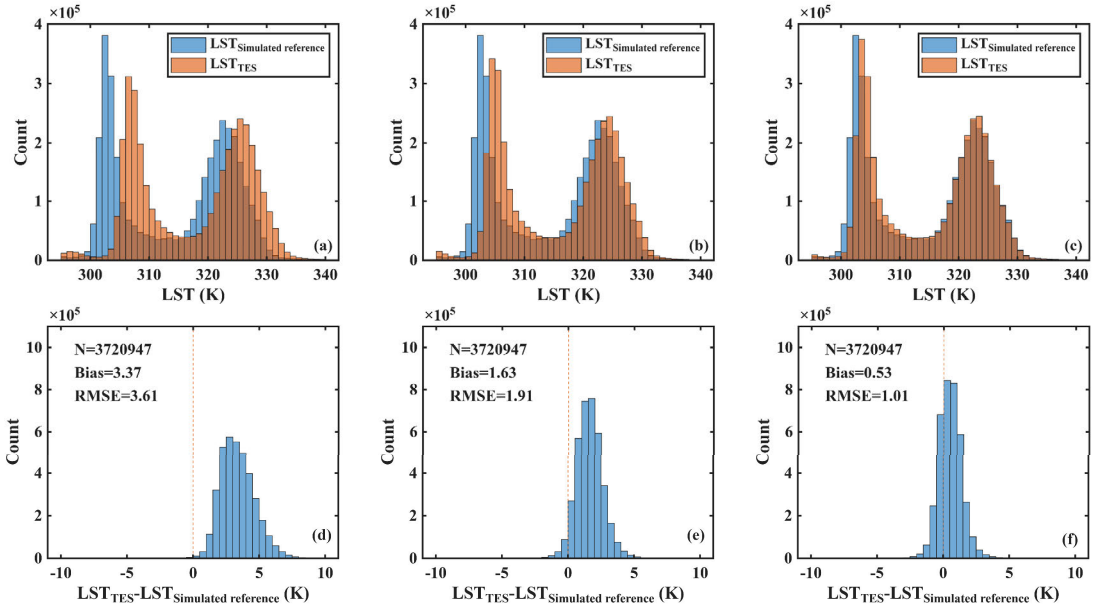


Fig. 16. Grayscale distributions of LST and their corresponding errors obtained by the three groups: (a) and (d) Group 1, (b) and (e) Group 2, and (c) and (f) Group 3.

reference values. This further demonstrates that the atmospheric parameters derived from SSDATER are more accurate and can improve the performance of the TES algorithm. Among all groups, the LST retrieved using the SSDATER method (i.e., Group 3) was the closest to the simulated reference LST.

Fig. 16 shows the grayscale distributions of LSTs of three groups, as well as the histograms of their deviations.

As shown in Fig. 16, the LSTs derived from both Group 1 and Group 2 exhibit significant overestimation in low LST regions, which correspond to water bodies and vegetation, which have low spectral contrast and are particularly sensitive to errors in atmospheric correction [Fig. 16(a) and (b)]. In contrast, the SSDATER method proposed in this study significantly reduced the bias and achieved the highest LST accuracy (RMSE = 1.01 K), which represented improvements of 2.6 and 0.9 K over those for Group 1 and Group 2, respectively.

Fig. 17 shows the spatial distribution of the simulated reference LSE for Hyper-CAM LW band 40, along with the LSE spatial distributions estimated by the three groups. As illustrated, the LSE estimated using the SSDATER method most closely matches the simulated reference values over both vegetation and water areas. In terms of spatial texture, the LSEs derived from Group 1 and Group 2 fail to effectively distinguish between vegetation and buildings due to high instrument noise. In contrast, the SSDATER method can clearly differentiate between vegetation and buildings.

Fig. 18 presents the estimated LSE spectra for the three groups across different LC types. Since Group 1 and Group 2 did not use the first six bands, the LSE values for these bands were calculated directly using the at-sensor radiance and the retrieved LST and are therefore excluded from the comparison.

As shown in Fig. 18, the mean LSE of Group 1 deviates the most from the simulated reference value and has the largest standard deviation for all LC types, while the second group performs slightly better. The mean LSE of Group 3 is closest to the simulated reference value and has the smallest standard deviation. Fig. 19 shows the LSE errors of the three groups without distinguishing between LC types.

Fig. 19 shows that Group 1 had the lowest accuracy, with an average error (from bands 7 to 81) close to 0.06 (average RMSE = 0.053). The average LSE error of Group 2 was approximately 0.04 (average RMSE = 0.036), whereas the average error of Group 3 was less than 0.02 (average RMSE = 0.016).

The validation results demonstrate that the SSDATER method proposed in this study can estimate more accurate atmospheric parameters from Hyper-CAM LW hyperspectral at-sensor radiance. Moreover, it enables the retrieval of high-precision LST and LSE even in the presence of instrument noise.

To comprehensively evaluate the reliability and precision of the proposed SSDATER framework, we conducted extensive supplementary analyses addressing its foundational assumptions, denoising efficacy, and sensitivity characteristics. Statistical analysis of 1662 surface spectra verifies the rationality of the spectral aggregation strategy. The mean correlation coefficient between bands remains above 0.95 at a lag of 10. Performance benchmarking under various atmospheric and surface conditions indicates that under the present experimental setup, SSDATER achieved lower errors than the tested off-the-shelf comparison methods, maintaining remarkable structural fidelity and low radiometric bias across all 81 bands. These quantitative findings confirm the method's high accuracy and stability in LST&E retrieval under complex noise distributions. Due to space constraints, the detailed datasets, quantitative comparative figures, sensitivity test, and extended discussions on mineral feature extraction are provided in the supplementary material.

V. Discussion

A. SSDATER Method in Support of PSF Effect Correction

Recent studies have shown that, according to both measured and simulated PSFs of sensor systems, a significant portion of

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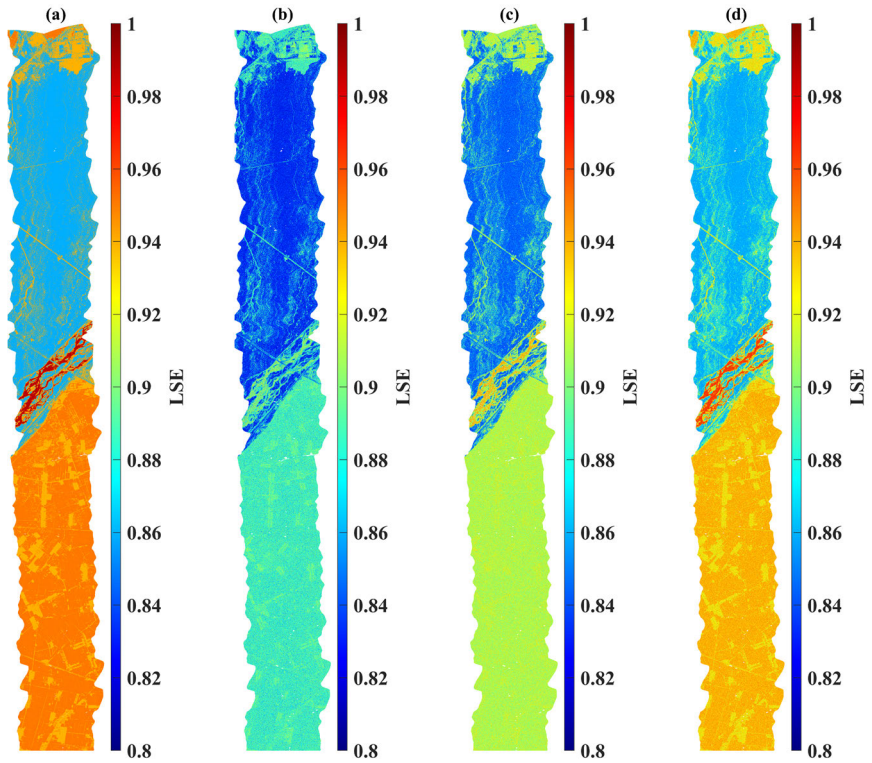


Fig. 17. Spatial distributions of the simulated reference LSE (a), and LSEs retrieved from the three groups: (b) Group 1, (c) Group 2, and (d) Group 3.

the signal detected by each pixel in remote sensing imagery actually comes from regions outside the nominal footprint of that pixel [48]. This effect limits the accuracy of the retrieved surface parameters at the individual pixel level. Research has indicated that spatial deconvolution methods can mitigate the influence of neighboring pixel energy, thereby enhancing the accuracy of LC classification and the inversion of surface parameters [45].

However, spatial deconvolution, especially when noise is present, can often amplify the noise component, which sometimes makes the results worse than those from the original blurred image [55], [56]. The SSDATER method can effectively reduce instrument noise. This raises the question of whether it can further support subsequent spatial deconvolution to eliminate the energy contributions from neighboring pixels, thereby enabling high-accuracy LST&E retrieval.

To investigate this, we assume that the PSF of Hyper-CAM LW is known beforehand. The Lucy–Richardson algorithm was then separately applied to the 81 hyperspectral bands and the nine synthesized multispectral bands for spatial deconvolution [57], [58]. Subsequently, the TES algorithm was performed separately on the spatially deconvolved at-sensor radiance for both data types.

Using the at-sensor brightness temperature under theoretical conditions (i.e., excluding instrument noise and point spread effects) as a simulated reference, Fig. 20 shows the errors in the at-sensor brightness temperature for both the 81 hyperspectral bands and the nine synthesized multispectral bands, before and after spatial deconvolution.

As shown in Fig. 20, direct spatial deconvolution of the 81 hyperspectral bands amplified these errors in all the bands beyond the instrument noise level. In contrast, while the nine synthesized multispectral bands initially exhibited greater errors than the instrument noise, these errors were reduced below the noise level after deconvolution. Spatial deconvolution may introduce additional errors in high-noise bands, but the proposed SSDATER method effectively reduces noise, thereby leading to postdeconvolution errors in the at-sensor radiance of the nine synthesized multispectral bands being lower than the instrument noise [Fig. 20(b)].

After spatial deconvolution, atmospheric correction was applied to both the 81 spatially deconvolved hyperspectral bands and the nine synthesized multispectral bands with atmospheric parameters estimated by the SSDATER method. Subsequently, the TES algorithm was used to retrieve LST&E for each dataset. The accuracies of the LSTs retrieved from the two datasets are shown in Fig. 21.

As shown in Fig. 21, after spatial deconvolution, the accuracy of LST retrieval from the 81 hyperspectral bands decreased, with the RMSE increasing from 1.91 to 3.15 K, primarily because the relatively high instrument noise in the hyperspectral data was amplified by spatial deconvolution. In contrast, the LST retrieval error decreased after spatial deconvolution for the nine synthesized multispectral bands, with the RMSE decreasing from 1.01 to 0.91 K. In summary, the proposed SSDATER can reduce instrument noise, further support the correction of PSF effects, and improve the accuracy of LST&E retrieval.

It should be noted that the spatial deconvolution assumes a known Gaussian PSF in this section. In practical applications, apart from prior knowledge of the intrinsic PSF characteristics of the sensor, additional PSF contributions induced by terrain

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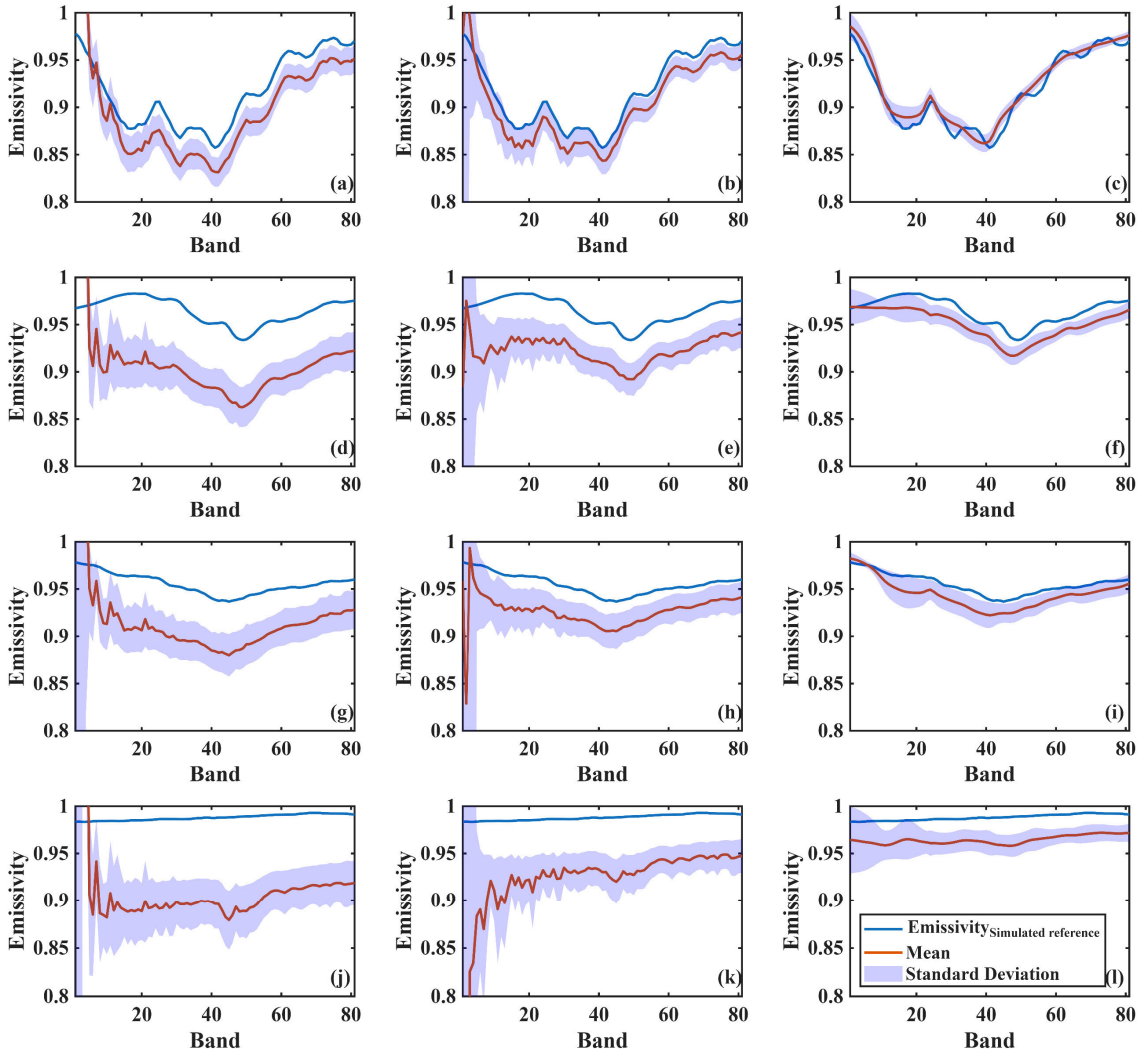


Fig. 18. Spectral plots that show the means and standard deviations of LSEs retrieved from three groups (Group 1: first column; Group 2: second column; and Group 3: third column) for various LC types: (a)–(c) barren land, (d)–(f) cropland, (g)–(i) built-up areas, and (j)–(l) rivers.

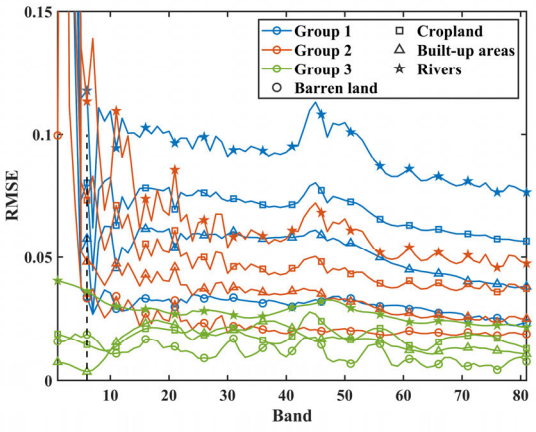


Fig. 19. Errors of LSEs retrieved from different groups at different bands.

and atmospheric effects also need to be taken into account. Consequently, relevant spatial deconvolution results are only exploratory preliminary findings.

B. Effects of Spatial and Spectral Aggregation Window Sizes on the Performance of SSDATER Method

The SSDATER method employs two key strategies—spatial aggregation and spectral aggregation—to effectively reduce instrument noise. These strategies facilitate accurate estimation of atmospheric parameters and enhance the precision of LST&E retrievals. The spatial aggregation window size is set to 20, while the spectral aggregation window size is set to 9, meaning that nine hyperspectral bands are synthesized into a multispectral band. This section mainly discusses the rationale behind these two default settings.

First, for spatial aggregation, we set window sizes of 5, 10, 20, 30, and 50 to perform unsupervised classification and spatial aggregation, respectively. We also aggregated the simulated reference at-sensor brightness temperature accordingly, assessing the accuracy of spatial aggregation at different window sizes. As shown in Fig. 22, the accuracy is lowest when the window size is set to 5. A possible reason for this is that the window size is too small, resulting in too few pixels participating in spatial aggregation, which leads to ineffective noise reduction. Although the accuracy of the aggregated

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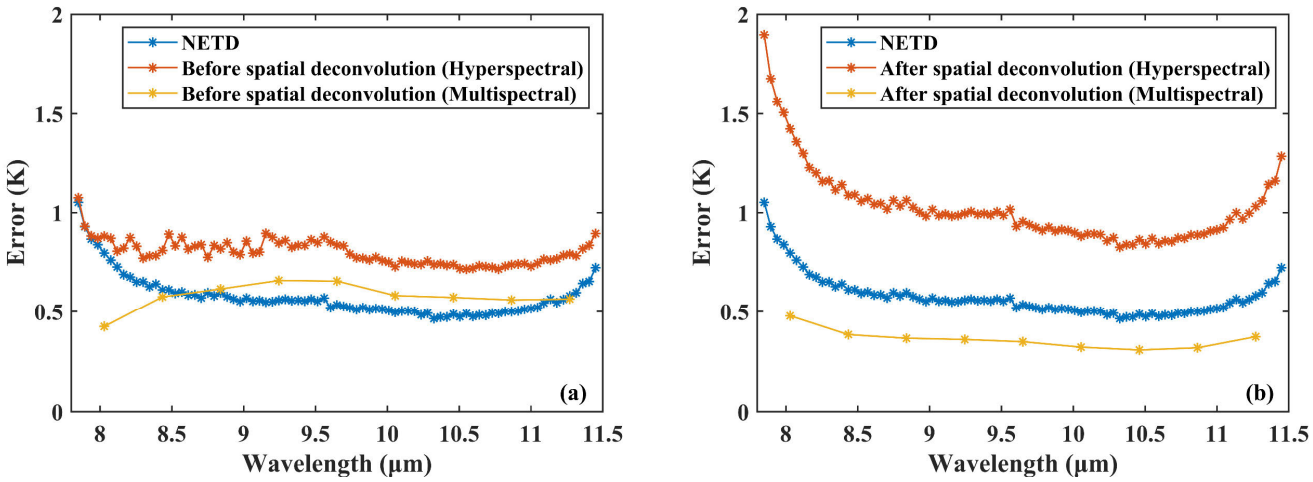


Fig. 20. Errors between the at-sensor hyperspectral and multispectral brightness temperatures before spatial deconvolution (a) and after spatial deconvolution (b).

TABLE IV
NUMBER OF PIXELS OBTAINED AFTER SPATIAL AGGREGATION AT DIFFERENT WINDOW SIZES UNDER DIFFERENT ATMOSPHERIC CONDITIONS

Window size	Tropical	Mid-latitude summer	Mid-latitude winter	Subarctic summer	Subarctic winter
N=5	11715	6041	5513	3943	5662
N=10	6820	4035	3501	2510	3609
N=20	1322	742	611	452	682
N=30	325	154	104	69	117
N=50	19	4	1	0	2

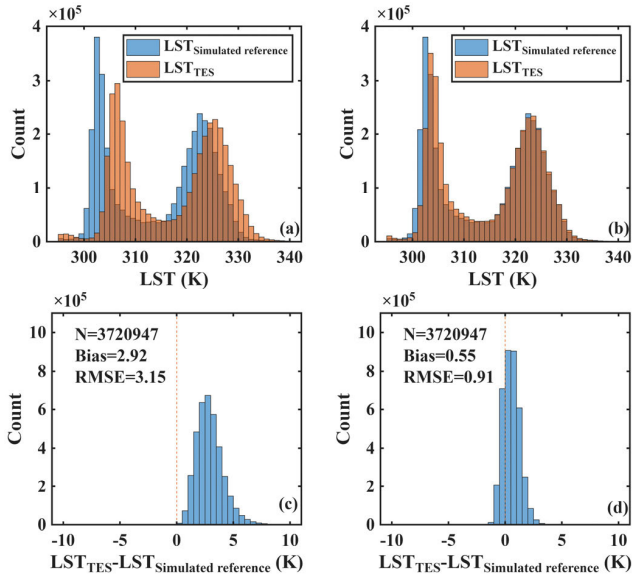


Fig. 21. Grayscale distributions of LSTs estimated from at-sensor radiance after spatial deconvolution and the corresponding error distributions: (a) and (c) 81 hyperspectral bands and (b) and (d) nine synthesized multispectral bands.

at-sensor brightness temperature generally improves as the window size increases, it does not necessarily continue to rise when the window becomes excessively large (e.g., N = 50), as shown in Fig. 22(e). This reflects a clear tradeoff: smaller windows provide insufficient noise reduction, while larger windows risk introducing excessive surface heterogeneity that increases the standard deviation of the at-sensor brightness temperature corresponding to dominant LC types from unsupervised classification. This makes it increasingly difficult for the dominant class to meet the criterion that the standard deviation across all 81 bands is less than 1 K, and may even cause the entire image to fail the requirements for spatial aggregation [see Fig. 22(d)]. Table IV presents the number of pixels obtained after spatial aggregation at various window sizes. It indicates that as the window size increases, the total number of aggregated pixels gradually decreases, and may even drop to zero under subarctic summer conditions when N = 50.

Overall, based on the results from Fig. 22 and Table IV, the window size used in this study (N = 20) is deemed appropriate, as it achieves a good balance between effective noise suppression and maintaining sufficient valid aggregated pixels.

In addition, we established various spectral aggregation strategies (convolving N hyperspectral channels to form a multispectral channel) and evaluated the effectiveness of each aggregation strategy based on the accuracy of LST retrieval. Table V lists the different spectral aggregation strategies. As shown in Table V, as N increases, the accuracy of LST retrieval also improves. The average LST retrieval accuracy improves by 0.3 K when comparing N = 9 to N = 6. However,

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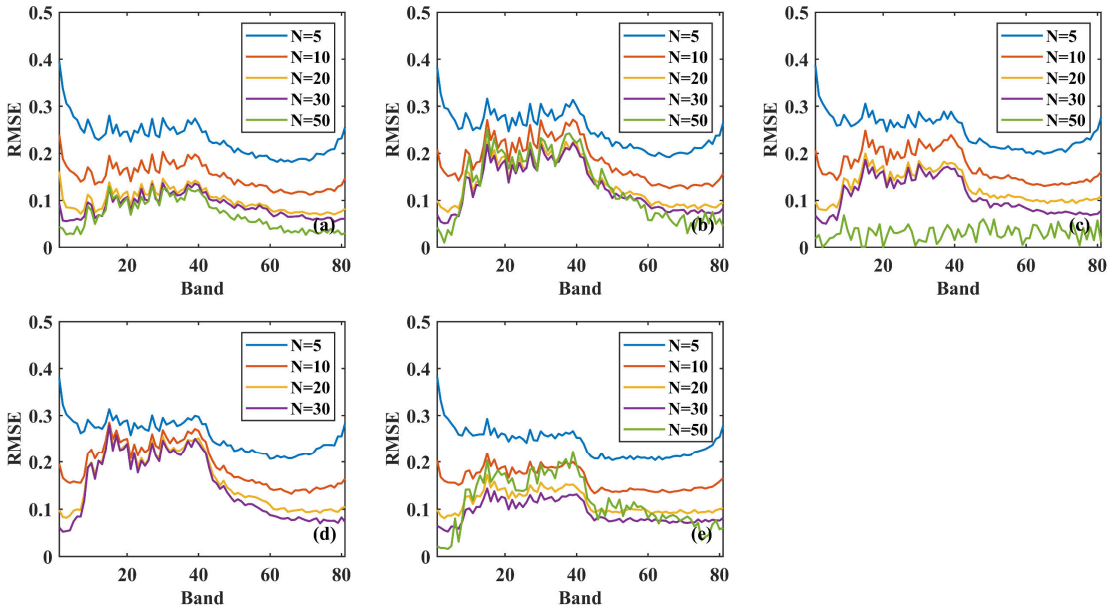


Fig. 22. Comparison of accuracy for at-sensor brightness temperatures after aggregation with different window sizes under different atmospheric conditions. (a) Tropical, (b) mid-latitude summer, (c) mid-latitude winter, (d) subarctic summer, (e) subarctic winter.

TABLE V
COMPARISON OF LST RETRIEVAL ACCURACY UNDER DIFFERENT ATMOSPHERIC CONDITIONS AND VARIOUS SPECTRAL AGGREGATION STRATEGIES

Number of bands	Tropical	Mid-latitude summer	Mid-latitude winter	Subarctic summer	Subarctic winter	Mean (K)
N=6	1.85	1.68	1.00	1.21	0.95	1.34
N=9	1.31	1.35	0.80	0.86	0.79	1.02
N=12	1.38	1.14	0.77	0.97	0.78	1.01
N=15	1.39	1.12	0.76	0.94	0.77	1.00

the improvement in LST retrieval accuracy from $N = 12$ and $N = 15$ compared to $N = 9$ is limited (less than 0.05 K). As such, the window size $N = 9$ adopted in this study serves as a balanced practical compromise, which jointly optimizes noise suppression, LST retrieval accuracy, preservation of spectral features, and the reconstruction stability of PCA.

C. Merits and Demerits of SSDATER

The SSDATER method mitigates sensor observation noise through spatial aggregation and spectral aggregation, two complementary strategies, and offers the following key advantages.

1) SSDATER performs local unsupervised classification and spatial aggregation of homogeneous pixels, thereby enhancing the at-sensor radiance without compromising spectral resolution. This improvement supports the estimation of atmospheric parameters under emissivity smoothness constraints and effectively addresses the shortcomings of traditional methods such as AAC and ISAC, which rely solely on airborne hyperspectral data and often fail to accurately retrieve atmospheric downwelling radiance from image-intrinsic information.

2) SSDATER leverages the inherent redundancy in hyperspectral emissivity to reduce noise through spectral aggregation while maintaining spatial resolution. This strategy helps enhance image clarity while retaining reliable at-sensor radiance measurements, which in turn contributes to more accurate temperature–emissivity separation results.

However, SSDATER still has certain limitations.

1) When estimating LST&E, it continues to rely on the classical TES algorithm. This approach does not incorporate targeted radiative modeling for complex surfaces such as urban areas or mountainous regions with significant topographic variation. As a result, its applicability is limited in these contexts. Meanwhile, airborne TIR hyperspectral sensors are often employed in applications such as urban heat island monitoring, where accurate temperature–emissivity separation is particularly critical.

2) The denoising in SSDATER is designed primarily for additive Gaussian noise, while other types of noise (such as stripe noise) are not adequately addressed.

3) When searching for the optimal atmospheric solution, the SSDATER framework extracts candidate atmospheric profiles from the TUD database guided by reanalysis background fields. Low accuracy in these reanalysis background profiles may therefore degrade the precision of atmospheric parameter retrievals produced by SSDATER, and the inherent completeness of the TUD database constitutes an additional influencing factor. Beyond that, surface materials bearing sharp emissivity

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spectral features introduce abrupt spectral fluctuations that conflict with the smoothness prior assumption; this compromises the robustness of the screening procedure and interferes with the voting-based identification of the optimal atmospheric profile.

4) The SSDATER method is applied to Hyper-CAM LW data that are affected by compounded noise, particularly in the 7.8–8.5 μm and 11–11.5 μm spectral ranges, where atmospheric water vapor/ozone absorption and instrumental noise jointly degrade the radiometric quality. Although the current method can partly mitigate these effects through denoising and atmospheric parameter retrieval, it does not explicitly model the band-dependent interaction between atmospheric and instrumental noise sources. As a result, these bands remain more challenging and may contribute disproportionately to retrieval uncertainty, especially under conditions with strong water vapor variability or suboptimal instrumental performance. This limitation indicates the need for more sophisticated noise-handling strategies, such as explicit band-dependent noise modeling, adaptive weighting schemes, or joint inversion of atmospheric and instrumental error components. Incorporating such mechanisms in future work is expected to improve the robustness and reliability of the retrievals, particularly in spectrally complex regions where compounded noise plays a dominant role.

VI. Conclusion

Airborne TIR hyperspectral sensors can acquire not only high spatial resolution LST data but also LSE data with both high spatial resolution and high spectral resolution. These high-quality datasets provide strong support for a wide range of applications, including geological exploration, urban thermal environment studies, agricultural remote sensing, and disaster assessment. However, such sensors often experience greater instrument noise due to platform stability, integration time, background interference, high spatial–spectral resolution, etc.

To address this issue, this study proposes an SSDATER method, which can effectively reduce sensor noise while preserving the physical meaning of the sensor's observational data. Furthermore, it can make full use of the spatial and spectral information in airborne TIR hyperspectral observation data to obtain high-precision atmospheric parameters and LST&E data.

The proposed SSDATER method was evaluated using simulated airborne TIR Hyper-CAM LW data. The results show that: 1) the atmospheric parameters retrieved using the SSDATER method were more accurate than those derived from reanalysis profiles (e.g., MERRA2) and those estimated by the ESSC algorithm and 2) under identical atmospheric correction conditions, the SSDATER method achieved the highest accuracy for both LST and LSE, with an LST error of 1.01 K and an average LSE error of 0.016. The LST and LSE directly estimated using the TES algorithm from the original Hyper-CAM LW data had an LST error of 1.91 K and an average LSE error of 0.036. In summary, the SSDATER method proposed in this study can retrieve high-precision LST and LSE from noisy airborne TIR hyperspectral data, thus supporting subsequent geological and climatic studies. Nevertheless, because the present validation was based entirely on simulated Hyper-CAM LW data, we will further evaluate SSDATER using real airborne TIR hyperspectral observations to assess its robustness and practical applicability.

Appendix

After atmospheric correction using the at-sensor radiance and atmospheric parameters of the synthetic multispectral bands, the resulting BOA radiance [i.e., (A.2)] and the true BOA radiance [i.e., (A.1)] exhibit differences due to spectral sampling effects

$$BOA_{true}(b) = \int_{\lambda_1}^{\lambda_2} \frac{L_{TOA}(\lambda) - L_{\uparrow}(\lambda)}{\tau(\lambda)} \cdot f(\lambda) d\lambda \tag{A.1}$$

$$BOA_{\text{est}}(b) = \frac{\int_{\lambda_1}^{\lambda_2} L_{TOA}(\lambda) \cdot f(\lambda) d\lambda - \int_{\lambda_1}^{\lambda_2} L_{\uparrow}(\lambda) \cdot f(\lambda) d\lambda}{\int_{\lambda_1}^{\lambda_2} \tau(\lambda) \cdot f(\lambda) d\lambda} = \frac{L_{TOA}(b) - L_{\uparrow}(b)}{\tau(b)} \tag{A.2}$$

where the $L_{TOA}(\lambda)$, $L_{\uparrow}(\lambda)$, and $\tau(\lambda)$ are the spectral at-sensor radiance, atmospheric upward radiance and transmittance, respectively. While the $f(\lambda)$ is the normalized spectral response function (i.e., $\int_{\lambda_1}^{\lambda_2} f(\lambda) d\lambda = 1$). The λ_1 and λ_2 are the lower and upper limits of the band b, respectively. The $L_{TOA}(b)$, $L_{\uparrow}(b)$ and $\tau(b)$ are the at-sensor radiance, atmospheric upward radiance and transmittance of band b, respectively.

Equation (A.1) is the theoretical truth, whereas (A.2) represents the outcome following atmospheric correction in a realistic environment. A deviation from (A.1) is therefore unavoidable, particularly in spectral bands characterized by substantial differences in atmospheric absorption or emissivity. Since $f(\lambda)$ does not vary with atmospheric and surface conditions, we treat it as a constant and perform a multivariate Taylor expansion on (A.1). For the sake of brevity, we will use x, y and z to refer to the variables above

$$g(x, y, z) = \frac{x - y}{z} \tag{A.3}$$

where $x = L_{TOA}(\lambda)$, $y = L_{\uparrow}(\lambda)$, $z = \tau(\lambda)$. Then, perform a Taylor expansion of $g(x, y, z)$ around the point (x_0, y_0, z_0), and only retain terms up to the second order in the expansion

$$\begin{aligned} &g(x, y, z) \\ &\approx g(x_0, y_0, z_0) + \nabla g(x_0, y_0, z_0)(x - x_0, y - y_0, z - z_0) \\ &\quad + \frac{1}{2}(x - x_0, y - y_0, z - z_0)^T H(g)(x_0, y_0, z_0) \\ &\quad \times (x - x_0, y - y_0, z - z_0) \end{aligned} \tag{A.4}$$

where $\nabla g(x_0, y_0, z_0)$ denotes the vector of first-order partial derivatives of the function $g(x, y, z)$ at (x_0, y_0, z_0). While $H(g)(x_0, y_0, z_0)$ is the matrix of second-order partial derivatives of the function $g(x, y, z)$ at the point (x_0, y_0, z_0). And the x_0, y_0, z_0 can be written as

$$\begin{aligned} x_0 &= \int_{\lambda_1}^{\lambda_2} L_{TOA}(\lambda) \cdot f(\lambda) d\lambda = L_{TOA}(b) \\ y_0 &= \int_{\lambda_1}^{\lambda_2} L_{\uparrow}(\lambda) \cdot f(\lambda) d\lambda = L_{\uparrow}(b) \\ z_0 &= \int_{\lambda_1}^{\lambda_2} \tau(\lambda) \cdot f(\lambda) d\lambda = \tau(b). \end{aligned} \tag{A.5}$$

By substituting the respective terms into (A.4), and integrate (A.4) we can obtain (A.6), shown at the top of the next page.

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$$
\begin{aligned}
&BOA_{true}(b)\\
&= \int \left[\begin{array}{c} \dfrac{x_0-y_0}{z_0} + \dfrac{1}{z_0}(x-x_0) - \dfrac{1}{z_0}(y-y_0) - \dfrac{x_0-y_0}{z_0^2}(z-z_0) + \dfrac{x_0-y_0}{z_0^3}(z-z_0)^2 \\ -\dfrac{1}{z_0^2}(x-x_0)(z-z_0) + \dfrac{1}{z_0^2}(y-y_0)(z-z_0) + \cdots \end{array} \right] \cdot f(\lambda) d\lambda \\
&= BOA_{est}(b) + \int \left[\begin{array}{c} \dfrac{1}{\tau(b)}(L_{TOA}(\lambda) - L_{TOA}(b)) - \dfrac{1}{\tau(b)}(L_{\uparrow}(\lambda) - L_{\uparrow}(b)) - \dfrac{L_{TOA}(b) - L_{\uparrow}(b)}{\tau(b)^2}(\tau(\lambda) - \tau(b)) \\ + \dfrac{L_{TOA}(b) - L_{\uparrow}(b)}{\tau(b)^3}(\tau(\lambda) - \tau(b))^2 - \dfrac{1}{\tau(b)^2}(L_{TOA}(\lambda) - L_{TOA}(b))(\tau(\lambda) - \tau(b)) \\ + \dfrac{1}{\tau(b)^2}(L_{\uparrow}(\lambda) - L_{\uparrow}(b))(\tau(\lambda) - \tau(b)) + \cdots \end{array} \right] \cdot f(\lambda) d\lambda
\end{aligned} \tag{A.6}
$$

Considering the variation of the atmospheric parameters across the spectral band, we define for each parameter

$$
\begin{aligned}
\Delta L_{TOA}(\lambda) &= L_{TOA}(\lambda) - L_{TOA}(b)\\
\Delta L_{\uparrow}(\lambda) &= L_{\uparrow}(\lambda) - L_{\uparrow}(b)\\
\Delta \tau(\lambda) &= \tau(\lambda) - \tau(b).
\end{aligned} \tag{A.7}
$$

According to the definition, the band-weighted average for each of these variables in (A.7) is zero

$$
\begin{aligned}
&\int \Delta L_{TOA}(\lambda) \cdot f(\lambda) d\lambda\\
&\quad = \int (L_{TOA}(\lambda) - L_{TOA}(b)) \cdot f(\lambda) d\lambda\\
&\quad = \int L_{TOA}(\lambda) \cdot f(\lambda) d\lambda - \int L_{TOA}(b) \cdot f(\lambda) d\lambda\\
&\quad = L_{TOA}(b) - L_{TOA}(b)\\
&\quad = 0.
\end{aligned} \tag{A.8}
$$

Similarly, $\int \Delta L_{\uparrow}(\lambda) \cdot f(\lambda) d\lambda = 0$, and $\int \Delta \tau(\lambda) \cdot f(\lambda) d\lambda = 0$. Therefore, the integral of the first-order term in (A.6) is 0. Ultimately, the discrepancy between the true and estimated BOA values is given by

$$
\begin{aligned}
&BOA_{true}(b)\\
&\approx BOA_{est}(b) + \frac{L_{TOA}(b) - L_{\uparrow}(b)}{\tau(b)^3} \int (\tau(\lambda) - \tau(b))^2 \cdot f(\lambda) d\lambda\\
&\quad + \frac{1}{\tau(b)^2} \int (L_{\uparrow}(\lambda) - L_{\uparrow}(b) - L_{TOA}(\lambda)\\
&\quad + L_{TOA}(b)) \cdot (\tau(\lambda) - \tau(b)) \cdot f(\lambda) d\lambda.
\end{aligned} \tag{A.9}
$$

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